

In-Depth Configuration Topics – Radio

AP Default IP address – 192.168.0.1/24

SM Default IP address – 192.168.0.2/24

Backdoor IP – 169.254.1.1/24

Username – admin, Password – admin



ePMP
FORCE 300

Get Cost-Effective High Speed Connectivity with the
High-gain Force 300-25 802.11AC Wave 2 solution

Radio Configuration – Access Point

Radio Mode: This parameter controls the function of the device. All ePMP devices may be configured to operate as an Access Point (AP), Subscriber Module (SM). For initial link bring-up, choose Access Point.

Driver Mode: This parameter controls the mode of operation of the Access Point – An AP may be configured to operate in TDD PTP mode, or as an ePTP Master. For initial link bring-up, choose TDD. When the AP is an ePTP Master, the system is able to provide significantly lower latency than other modes.

Country – select country code from the drop down list.

Recommendation: India

SSID – Should be unique for each AP. Used to identify the AP identity from SM side

Max Range – This parameter represents link coverage distance. SM outside the configured range will not be able to connect.

Range Unit – should be same at both ends for better link distance calculation by radio.

Channel Selection – Can be manual by checking spectrum or we can use ACS or DCS to find out the clean channel.

Recommendation: Manual

Configuration > Radio

General

Radio Mode *

Driver Mode

Point-To-Point Access

Country

☒ Access Point

☐ Subscriber Module

☐ ePTP Master

☒ TDD PTP

☒ First Subscriber Module

☐ MAC Filtering

India

Access Point Configuration

SSID

Max Range

Range Unit

Channel Selection

Smallcell52177

8

☐ Miles

☒ Kilometers

☒ Manual

☐ DCS

☐ ACS

Radio Configuration – Access Point

TDD Ratio : we can choose different TDD ratios for uplink & downlink using scheduler. **Recommendation: 75/25 (DL/UL) – For Small cell**

Tx power & Antenna Gain: Antenna Gain is Fixed (25dBi) & Choose Tx Power as per EIRP requirement. **Recommendation: Use TX power as per Airtel guidelines – For Small cell**

Subscriber Module Target Receive Level:

This defines the desired receive power level at the AP from registered SMs. APs use this parameter to control the transmission power of their SMs in order to reduce system self-interference. **Recommendation: -40**

Channel Bandwidth: 20/40 & 80 MHz available.

Recommendation: 40MHz – For Small cell

Frequency: as per Spectrum Analyser Result.

Downlink Max Rate: Configure the MCS (Modulation and Coding Scheme) rate beyond which the radio's scheduler should not exceed when transmitting data traffic on the downlink.

This is useful in situations where there is high variance and unpredictability in the interference present in the environment causing packet loss. Reducing the max rate to a lower MCS (than the default DS9) may help in these situations.

Reducing the Downlink Max Rate will result in reduced link capacity.

Scheduler

Downlink/Uplink Ratio

70/30

60/40

50/50

40/60

30/70

-

75/25

65/35

55/45

45/55

35/65

+

Frame Size

5 ms

i

 Guard Interval

☐ Long

☒ Short

Power Control

i

 Transmitter Output Power

5

dBm | min: -9 | max: 30

i

 Antenna Gain

25

dBi | min: 25 | max: 40

i

 Subscriber Module Target Receive Level

-40

dBm | min: -80 | max: -40

Radio Configuration

Channel Bandwidth

☐ 80 MHz

☐ 40 MHz

☒ 20 MHz

Frequency Carrier

5560 MHz

i

 Downlink Max Rate

DS MCS 9 - 256-QAM 5/6

©2020 Cambium Networks, Ltd

Radio Configuration – Subscriber Module

Radio Mode: For initial link bring-up, choose Access Point.

Driver Mode: For initial link bring-up, choose TDD.

Preferred AP: Select 'add new AP' and configure:
SSID – same as Access Point.
Wireless Security Type – Select WPA2
Pre-shared Key - same as Access Point.

Tx power & Antenna Gain: Tx power should be auto and configure antenna gain as 25dBi.

Network Entry RSSI & SNR Threshold:

This defines the minimum value of RSSI & SNR on which RF link will connect successfully.

Channel Bandwidth: 20/40 & 80 MHz available.

Uplink Max Rate: Configure the MCS (Modulation and Coding Scheme) rate beyond which the radio's scheduler should not exceed when transmitting data traffic on the uplink.

Downlink Max Rate: Auto (As per access point configuration).

Frequency Scan List: Select all frequencies in 20,40 and 80MHz Channel bandwidth.

General

Radio Mode *

☐ Access Point
☒ Subscriber Module

Driver Mode

☒ TDD
☐ ePTP Slave

Preferred APs

Add new AP

Show Details

	SSID	Wireless Security	WPA2 Pre-shared Key
↑ ↓ ×	Smallcell52177	WPA2	

Power Control

Transmitter Output Power

15

dBm | min: 3 | max: 30

Antenna Gain

25

dBi | min: 25 | max: 40

Network Entry RSSI Threshold

-90

dBm | min: -100 | max: -20

Network Entry SNR Threshold

0

dB | min: -5 | max: 60

Radio Configuration

Max Tx Power

☒ Auto
☐ Manual

Uplink Max Rate

DS MCS 9 - 256-QAM 5/6

Downlink Max Rate

Use AP's config

Scan Channel Bandwidth

☒ 80 MHz
☒ 40 MHz
☒ 20 MHz

199 80 MHz Scan List

207 40 MHz Scan List

211 20 MHz Scan List

Quick Start



ePMP
FORCE 300

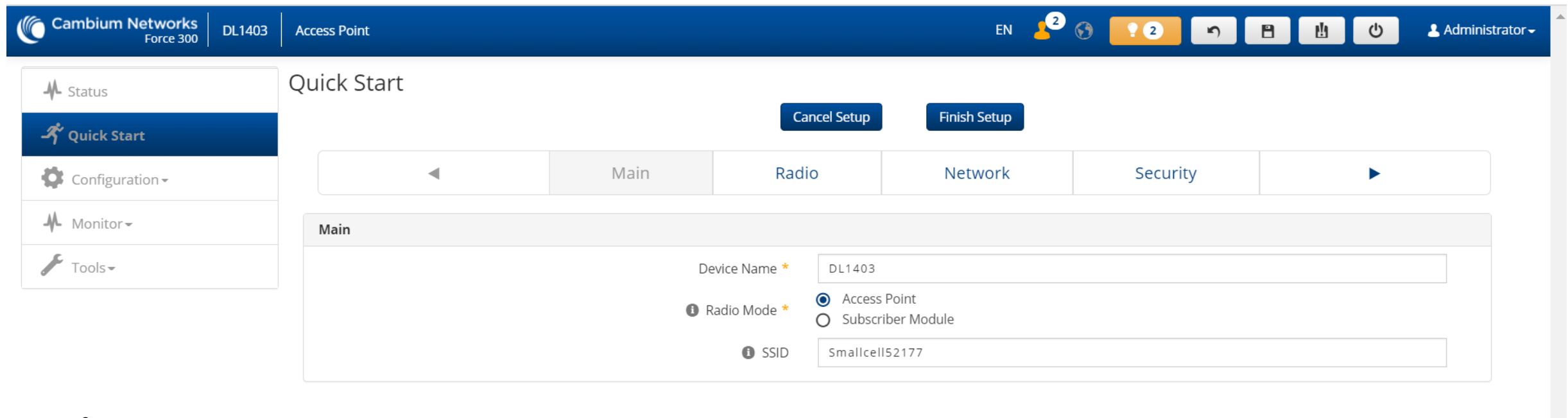
Get Cost-Effective High Speed Connectivity with the
High-gain Force 300-25 802.11AC Wave 2 solution

AP Quick Start

The Quick Start tab contains a listing of parameters required to configure a simple radio link and to configure requisite networking parameters. After configuring these parameters on the AP and SM and resetting both devices, the SM will be ready to associate (register) to the AP.

To configure an AP via the Quick Start, perform the following procedure:

1. Start the web browser from the management PC.
2. Navigate to menu **Quick Start**



The screenshot shows the Cambium Networks web interface for device DL1403. The top navigation bar includes the Cambium Networks logo, the device name 'DL1403', and the 'Access Point' tab. On the right side of the bar are icons for language (EN), notifications (2), a globe, a lightbulb (2), a refresh button, a save button, a print button, a power button, and a user profile labeled 'Administrator'. A left sidebar contains menu items: Status, Quick Start (highlighted), Configuration, Monitor, and Tools. The main content area is titled 'Quick Start' and features a 'Cancel Setup' and 'Finish Setup' button pair. Below these is a horizontal tab bar with 'Main', 'Radio', 'Network', and 'Security' tabs. The 'Main' tab is active, displaying configuration fields: 'Device Name' with the value 'DL1403', 'Radio Mode' with 'Access Point' selected (radio button) and 'Subscriber Module' as an option, and 'SSID' with the value 'Smallcell52177'.

Configure **Device Name**

Select Radio mode as **Access Point**

Configure a **Unique SSID**

AP Quick Start

Select Country as **India**

Select Driver mode as **TDD PTP**

Define **Max Range**

Channel Selection should be **Manual**

Select Channel BW as **40MHz**

Quick Start

Cancel Setup

Finish Setup



Main

Radio

Network

Security



Radio

 Country

India

 Driver Mode

☐ ePTP Master

☒ TDD PTP

 Max Range

4

kilometers | min: 1 | max: 200

 Channel Selection

☒ Manual

☐ DCS

☐ ACS

Channel Bandwidth

☐ 80 MHz

☒ 40 MHz

☐ 20 MHz

Frequency Carrier

5560 MHz

Select IP assignment as static & Configure IP, Subnet and Gateway

Quick Start

Cancel Setup

Finish Setup



Main

Radio

Network

Security



Network

IP Assignment ☒ Static ☐ DHCP

 IP Address 172.42.50.218

Subnet Mask 255.255.255.248

Gateway 172.42.50.217

Preferred DNS Server

Alternate DNS Server

Select Wireless security as **WPA-2** and enter a WPA-2 Pre-shared key

Quick Start

Cancel Setup

Finish Setup



Main

Radio

Network

Security



Security

 Wireless Security ☐ Open ☒ WPA2 ☐ RADIUS

 WPA2 Pre-shared Key

.....



Servers

Add

Show Details

IP Address	Port	Secret
------------	------	--------

Table is empty

 GUI User Authentication

- ☒ Device Local Only
☐ Remote RADIUS Server Only
☐ Remote RADIUS Server and Fallback to Local

Select Wireless security as **WPA-2** and enter a WPA-2 Pre-shared key

Quick Start

[Cancel Setup](#) [Finish Setup](#)

[◀](#) [Main](#) [Radio](#) [Network](#) [Security](#) [▶](#)

Security

Wireless Security

☐ Open ☒ WPA2 ☐ RADIUS

WPA2 Pre-shared Key



Servers

[Add](#) [Show Details](#)

IP Address	Port	Secret
Table is empty		

GUI User Authentication

☒ Device Local Only ☐ Remote RADIUS Server Only ☐ Remote RADIUS Server and Fallback to Local

Once all configuration is done, click on **Finish Setup** to save & apply all changes.

Configure **Device Name**

Select Radio mode as **Subscriber module**

Quick Start

Cancel Setup

Finish Setup



Main

Radio

Network

Security



Main

Device Name *

F300-25_ae74a7

 Radio Mode *



Access Point



Subscriber Module

Select Driver mode as **TDD**

Configure Same **SSID & Pre-shared Key** as configured in AP

Select all frequencies in 20/40/80MHz

Quick Start

Cancel Setup

Finish Setup

◀

Main

Radio

Network

Security

▶

Radio

Driver Mode

☒ TDD ☐ ePTP Slave

Preferred APs

Add new AP

Show Details

	SSID	Wireless Security	WPA2 Pre-shared Key
⬆ ⬇ ✕	Smallcell52177 ✎	WPA2 ✎	 ✎

◀ ▶

Scan Channel Bandwidth

✓ 80 MHz

✓ 40 MHz

✓ 20 MHz

199 80 MHz Scan List

207 40 MHz Scan List

211 20 MHz Scan List

Select IP assignment as static & Configure IP, Subnet and Gateway

Quick Start

Cancel Setup

Finish Setup

◀	Main	Radio	Network	Security	▶
---	------	-------	---------	----------	---

Network

i Network Mode ★ ☐ NAT ☒ Bridge ☐ Router

IP Assignment ☒ Static ☐ DHCP

i IP Address

Subnet Mask

Gateway

Preferred DNS Server

Alternate DNS Server

Select IP assignment as static & Configure IP, Subnet and Gateway

Quick Start

Cancel Setup

Finish Setup

◀

Main

Radio

Network

Security

▶

Network

Network Mode *

☐ NAT

☒ Bridge

☐ Router

IP Assignment

☒ Static

☐ DHCP

IP Address

Subnet Mask

Gateway

Preferred DNS Server

Alternate DNS Server

Once all configuration is done, click on **Finish Setup** to save & apply all changes.

In-Depth Configuration Topics – Quality of Service



ePMP
FORCE 300

Get Cost-Effective High Speed Connectivity with the
High-gain Force 300-25 802.11AC Wave 2 solution

Quality of Service Configuration

Traffic Priority : Enabled

VoIP Priority: Enabled

Add QoS classification rules for DSCP value 34,26,28 and 0.

DSCP 34 will have High priority.

Configuration ▶ Quality of Service

Maximum Information Rate (MIR)

MIR ☒ Disabled ☐ Enabled

MIR Profiles

Add

Show Details

	Number	Description	Downlink MIR (kbps)	Uplink MIR (kbps)
	0	default	1000000 	1000000 

Traffic Priority

 Traffic Priority ☐ Disabled ☒ Enabled

 VoIP Priority ☐ Disabled ☒ Enabled

 Broadcast Priority ☒ Low ☐ High

 Multicast Priority ☒ Low ☐ High

 QoS Classification Rules

Add

Show Details

	Type	Details	Priority
	CoS	5	Voice
	DSCP	46	Voice
  	DSCP 	34 	High 
  	DSCP 	26 	Low 
  	DSCP 	28 	Low 
  	DSCP 	0 	Low 

In-Depth Configuration Topics – System



ePMP
FORCE 300

Get Cost-Effective High Speed Connectivity with the
High-gain Force 300-25 802.11AC Wave 2 solution

System Configuration

Device Name: To identify the AP in a tower

Inactive logout Period can be set by the user

Web Access: HTTP & HTTPs for secure access

SSH /Telnet : SSH is enabled default where as telnet is disabled by default considering security

SNMP: Supports v1, v2c and v3

Trap Servers : Up to four trap servers can be configured

Syslog Server: Up to four syslog server can be configured

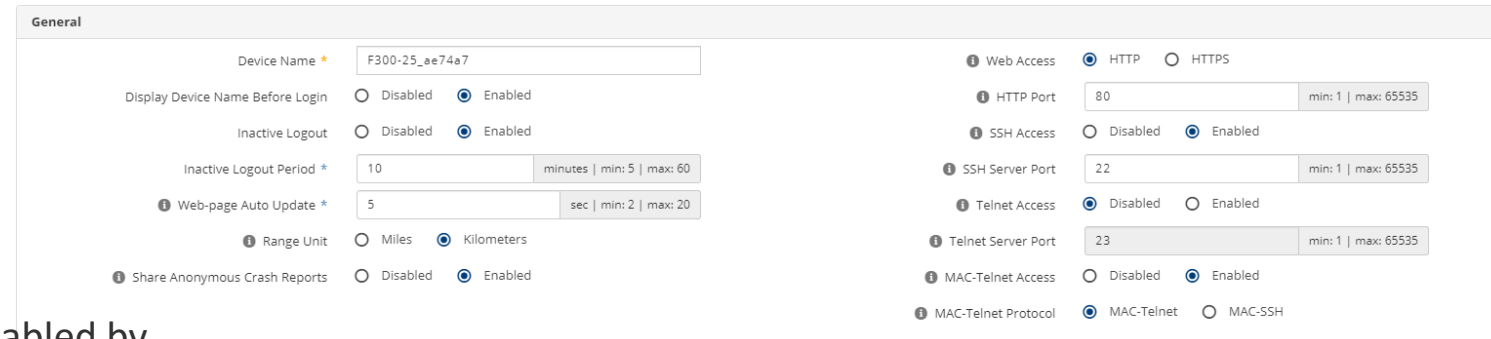
NTP Server: Accurate time through **NTP** is important for troubleshooting as Syslog entries are time stamped.

Static: The device retrieves NTP time data from the servers configured in fields **NTP Server IP Address**

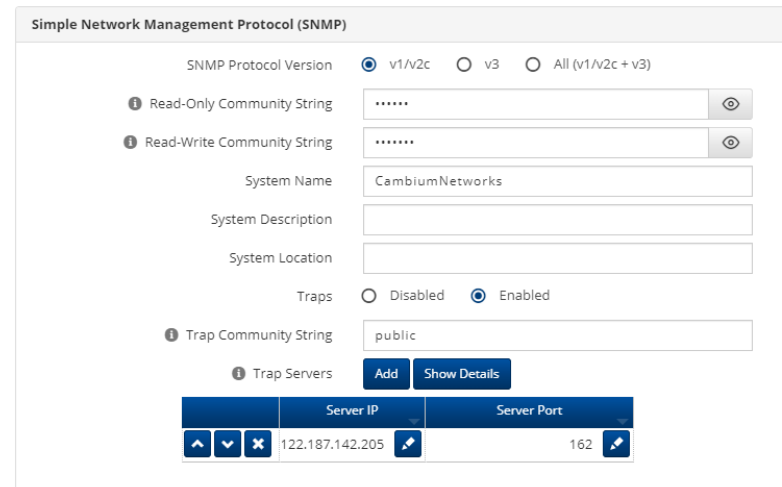
DHCP: The device retrieves NTP time data from the server IP issued via a network DHCP server

NTP Server 1 & 2: IP address of primary and secondary NTP server from which the device will retrieve time and date information.

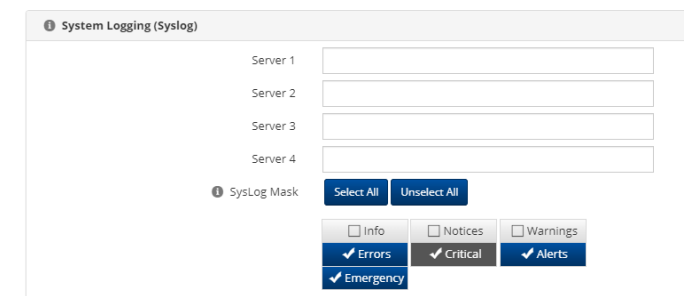
Time Zone: Set to operator local time zone or centralized time zone.



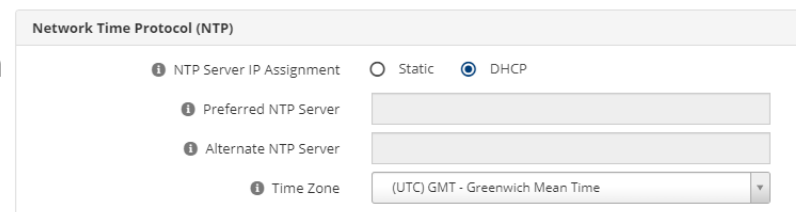
The General configuration page includes fields for Device Name (F300-25_ae74a7), Display Device Name Before Login (Enabled), Inactive Logout (Enabled), Inactive Logout Period (10 minutes), Web-page Auto Update (5 seconds), Range Unit (Kilometers), and Share Anonymous Crash Reports (Enabled). It also features checkboxes for Web Access (HTTP selected), SSH Access (Enabled), Telnet Access (Disabled), and MAC-Telnet Access (Enabled).



The SNMP configuration page shows the protocol version set to v1/v2c. It includes fields for Read-Only and Read-Write Community Strings, System Name (CambiumNetworks), System Description, System Location, and Traps (Enabled). A table for Trap Servers is shown with one entry: Server IP 122.187.142.205 and Server Port 162.



The System Logging (Syslog) configuration page includes fields for up to four servers. It also features a SysLog Mask section with checkboxes for Info, Notices, Warnings, Errors, Critical, and Alerts.

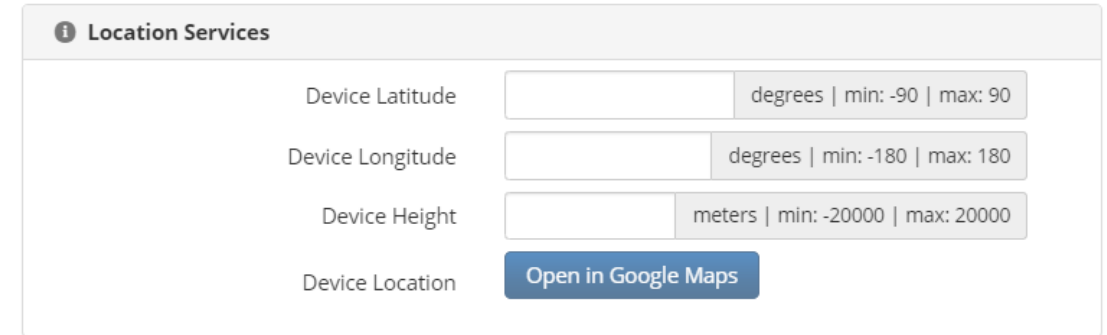


The Network Time Protocol (NTP) configuration page shows the NTP Server IP Assignment set to DHCP. It includes fields for Preferred NTP Server, Alternate NTP Server, and Time Zone (UTC - Greenwich Mean Time).

System Configuration

Location Services: Enter in the GPS location of the device. This is used by cnMaestro to place devices in the mapping display.

cnMaestro: Enter URL, Cambium ID and on-boarding Key. Device will be managed on cnMaestro for Monitoring & Troubleshooting



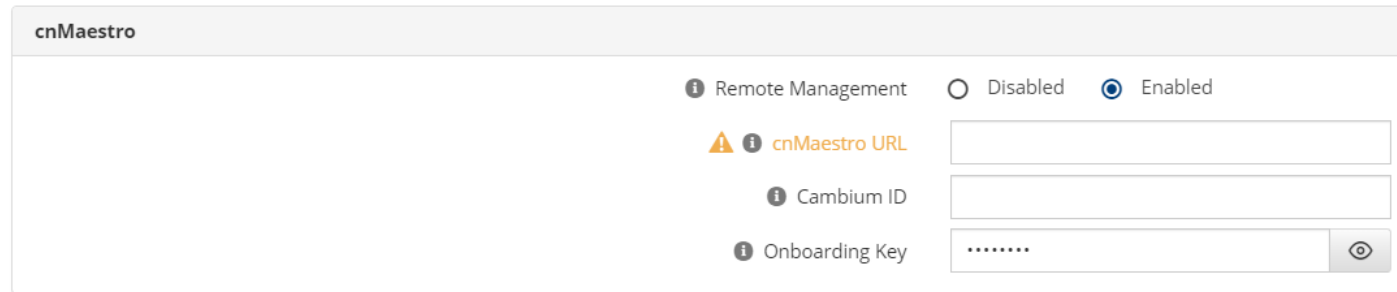
Location Services

Device Latitude degrees | min: -90 | max: 90

Device Longitude degrees | min: -180 | max: 180

Device Height meters | min: -20000 | max: 20000

Device Location [Open in Google Maps](#)



cnMaestro

Remote Management ☐ Disabled ☒ Enabled

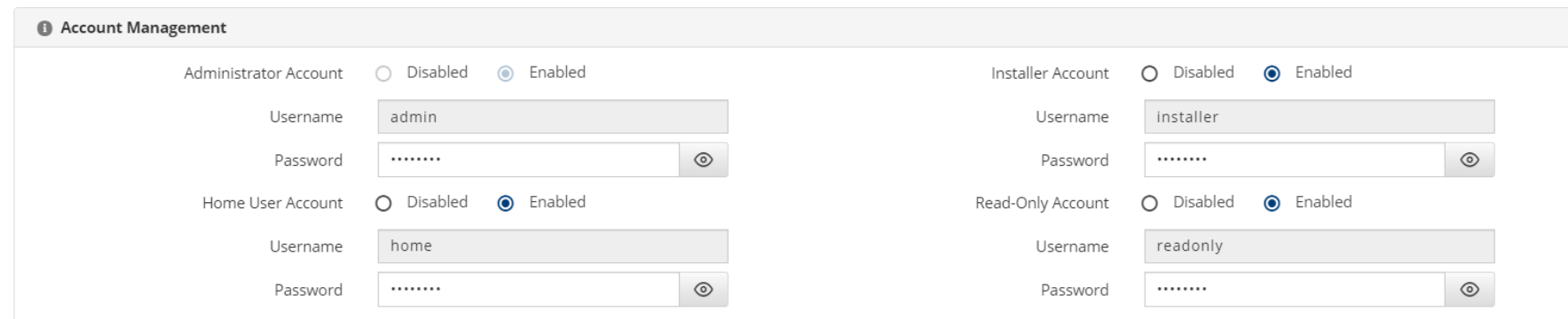
cnMaestro URL

Cambium ID

Onboarding Key

Account Management: There are four levels of Force 300 account management:

Administrator (admin)
Installer (installer)
Home User (home)
Read-Only (read only).



Account Management

Administrator Account ☐ Disabled ☒ Enabled

Username

Password

Home User Account ☐ Disabled ☒ Enabled

Username

Password

Installer Account ☐ Disabled ☒ Enabled

Username

Password

Read-Only Account ☐ Disabled ☒ Enabled

Username

Password

The account “Username” cannot be changed, but the “Password” can and should be changed.

The Administrator account can manage and disable the other accounts.

In-Depth Configuration Topics – Network



ePMP
FORCE 300

Get Cost-Effective High Speed Connectivity with the
High-gain Force 300-25 802.11AC Wave 2 solution

IP Address: It can be configured as:

Static – Manually enter IP settings

DHCP – IP Settings requested from DHCP Server

Note - if Device IP address Mode is set to DHCP and the device is unable to retrieve IP address information via DHCP, the device management IP is set to fallback IP 192.168.0.1 (AP mode), 192.168.0.2 (SM mode).

Units may always be accessed via the Ethernet port with IP 169.254.1.1

MTU: it is the size in bytes of the largest data unit that the device is configured to process. Larger MTU configurations can enable the network to operate with greater efficiency, but in the case of retransmissions due to packet errors, efficiency is reduced since large packets must be resent in the event of an error. Packets received by the device larger than the configured MTU will be dropped.

Default 1500, Max 1700

Ethernet Port Configuration:

Auto-Negotiate: AP auto negotiates the LAN Ethernet port speed and duplex mode with the device connected to it.

Manual: Manually set Ethernet port speed and duplex mode.

General

IP Assignment

☒ Static ☐ DHCP

IP Address

172.42.50.218

Subnet Mask

255.255.255.248

Gateway

172.42.50.217

Preferred DNS Server

Alternate DNS Server

Ethernet Port

Ethernet MTU *

1538

Port Setting

☒ Manual ☐ Auto-Negotiate

Port Speed

☐ 10 Mbps ☐ 100 Mbps ☒ 1000 Mbps

Port Duplex Mode

☐ Half ☒ Full

VLAN: Separates Management traffic (SNMP or HTTP/S) from user traffic.

Management interface can be accessed only by packets tagged with a VLAN ID matching the management VLAN ID.

MGMT VLAN ID: Enter VLAN ID (1-4095). Avoid using VLAN 1 as it often is implemented as “no VLAN”

MGMT VLAN Priority: set priority of management traffic (802.1p), values 0-4 (Low), 5-7 (High)

Management Access: should be Ethernet and Wireless, so that one can login the Access point from CPE end.

Ethernet: Only allow access to the AP’s management interface via a local Ethernet (LAN) connection.

Ethernet and Wireless: Allow access to the AP’s web management interface via a local Ethernet (LAN) connection and from over the air (i.e. from a device situated below the SM).

Virtual Local Area Network (VLAN)

Management VLAN

☐ Disabled

☒ Enabled

Management VLAN ID

560

min: 1 | max: 4094

Management VLAN Priority

1

min: 0 | max: 7

Advanced

IPv6 Support

☒ Disabled

☐ Enabled

Spanning Tree Protocol

☒ Disabled

☐ Enabled

Management Access

☐ Ethernet

☒ Ethernet And Wireless

DHCP Option 82

☒ Disabled

☐ Enabled

LLDP

☐ Disabled

☒ Enabled

LLDP Mode

☒ Receive and Transmit

☐ Receive only

PPPoE Intermediate Agent

☒ Disabled

☐ Enabled

Unicast packets with unknown MAC address

☐ Disabled

☒ Enabled

In-Depth Configuration Topics – Security



ePMP
FORCE 300

Get Cost-Effective High Speed Connectivity with the
High-gain Force 300-25 802.11AC Wave 2 solution

Open: All SMs requesting network entry are allowed registration.

WPA2: The WPA2 mechanism provides AES radio link encryption and SM network entry authentication. SMs must have the same Pre-shared Key as configured on the AP.

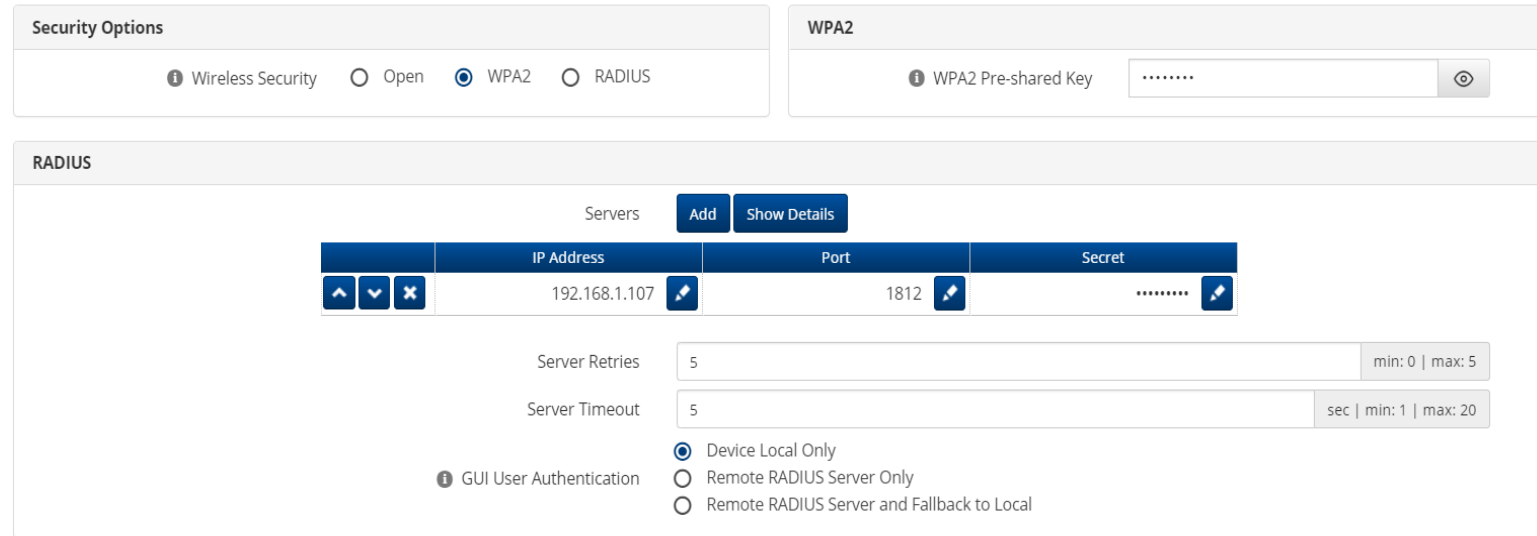
Pre-shared key must be between 8 to 128 symbols. Click the visibility icon  to toggle the display of the key's contents.

Radius: Authentication requests are sent to a Radius Server.

Radio Provides Radius authentication using Radius Server List (up to 3)

- IP Address of Server
- Port Address (Default 1812)
- Shared Secret matching Server
- Server Retry – retries per server in list before moving to next server
- Server Timeout – second to wait for a server response

Layer 2 & Layer 3 Firewall: Firewall rules can be defined to allow or deny a specific traffic on Wireless or Ethernet interface



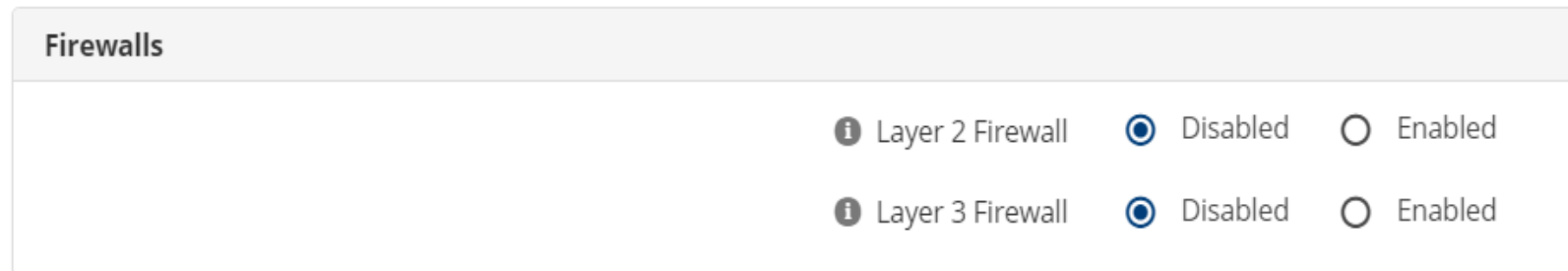
The screenshot displays the 'Security Options' section with three radio buttons: 'Wireless Security' (selected), 'Open', 'WPA2', and 'RADIUS'. Below this, the 'WPA2' section shows a 'WPA2 Pre-shared Key' field with a visibility icon. The 'RADIUS' section features a 'Servers' table with columns for 'IP Address', 'Port', and 'Secret'. A single server is listed with IP '192.168.1.107' and port '1812'. Below the table, 'Server Retries' is set to 5 and 'Server Timeout' is set to 5 seconds. At the bottom, 'GUI User Authentication' is set to 'Device Local Only'.

IP Address	Port	Secret
192.168.1.107	1812	

Server Retries: 5 (min: 0 | max: 5)
Server Timeout: 5 (sec | min: 1 | max: 20)

GUI User Authentication: ☒ Device Local Only, ☐ Remote RADIUS Server Only, ☐ Remote RADIUS Server and Fallback to Local

Recommendation: WPA-2 with Pre-shared key



The screenshot shows the 'Firewalls' section with two settings: 'Layer 2 Firewall' and 'Layer 3 Firewall'. Both are currently set to 'Disabled'.

Firewall Type	Status
Layer 2 Firewall	Disabled
Layer 3 Firewall	Disabled

Device Monitoring



ePMP
FORCE 300

Get Cost-Effective High Speed Connectivity with the
High-gain Force 300-25 802.11AC Wave 2 solution

Ethernet & Wireless Statistics: We can monitor Tx & Rx throughput & packet counters for ethernet as well as wireless interface.

Note: 'Retransmission Packet' in wireless statistics is a very critical parameter in terms of voice traffic. It should not exceed 2% for uninterrupted services.

System Statistics: Used to monitor Session Drops, Network entry Attempts & device reboot count.

Downlink & Uplink packet per MCS: We can check the packet percentage in each MCS value & can limit the MCS accordingly in order to avoid packet drops & retransmission.

Ethernet Statistics - Transmitted		
Total Traffic	796957.3 Kbytes (100%)	
Total Transmitted Packets	560563 packets (100%)	
Packet Errors	0 packets	
Packet Drops	0 packets	
Broadcast Packets	1604 packets (0%)	
Multicast Packets	361 packets (0%)	
Unicast packets	558599 packets (100%)	

Ethernet Statistics - Received		
Total Traffic	810778.6 Kbytes (100%)	
Total Received Packets	562109 packets (100%)	
Packet Errors	0 packets	
Packet Drops	0 packets	
Broadcast Packets	1407 packets (0%)	
Multicast Packets	2047 packets (0%)	
Unicast packets	558652 packets (99%)	

Wireless Statistics - Downlink		
Total Traffic	493821 Kbytes (100%)	
i Total Transmitted Packets	330188 packets (100%)	
Error Drop Packets	0 packets	
i Capacity Drop Packets	172 (0%)	
Retransmission Packets	16094 packets (5%)	
Multicast / Broadcast Traffic	50.9 Kbytes (0%)	
Broadcast Packets	358 packets (0%)	
Multicast Packets	89 packets (0%)	
Unicast packets	329741 packets (100%)	

Wireless Statistics - Uplink		
Total Traffic	173065.3 Kbytes (100%)	
i Total Received Packets	115558 packets (100%)	
Error Drop Packets	0 packets	
Multicast / Broadcast Traffic	1.9 Kbytes (0%)	
Broadcast Packets	35 packets (0%)	
Multicast Packets	0 packets	
Unicast packets	115523 packets (100%)	

i Downlink Packets Per MCS			
DS MCS 9 - 256-QAM 5/6	48496 (14%)	SS MCS 9 - 256-QAM 5/6	17 (0%)
DS MCS 8 - 256-QAM 3/4	237568 (68.6%)	SS MCS 8 - 256-QAM 3/4	1 (0%)
DS MCS 7 - 64-QAM 5/6	57073 (16.5%)	SS MCS 7 - 64-QAM 5/6	12 (0%)
DS MCS 6 - 64-QAM 3/4	2892 (0.8%)	SS MCS 6 - 64-QAM 3/4	11 (0%)
DS MCS 5 - 64-QAM 2/3	28 (0%)	SS MCS 5 - 64-QAM 2/3	4 (0%)
DS MCS 4 - 16-QAM 3/4	55 (0%)	SS MCS 4 - 16-QAM 3/4	4 (0%)
DS MCS 3 - 16-QAM 1/2	33 (0%)	SS MCS 3 - 16-QAM 1/2	1 (0%)
DS MCS 2 - QPSK 3/4	25 (0%)	SS MCS 2 - QPSK 3/4	23 (0%)
DS MCS 1 - QPSK 1/2	40 (0%)	SS MCS 1 - QPSK 1/2	12 (0%)

i Uplink Packets Per MCS			
DS MCS 9 - 256-QAM 5/6	68876 (90.1%)	SS MCS 9 - 256-QAM 5/6	45 (0.1%)
DS MCS 8 - 256-QAM 3/4	6585 (8.6%)	SS MCS 8 - 256-QAM 3/4	7 (0%)
DS MCS 7 - 64-QAM 5/6	194 (0.3%)	SS MCS 7 - 64-QAM 5/6	90 (0.1%)
DS MCS 6 - 64-QAM 3/4	70 (0.1%)	SS MCS 6 - 64-QAM 3/4	65 (0.1%)
DS MCS 5 - 64-QAM 2/3	78 (0.1%)	SS MCS 5 - 64-QAM 2/3	4 (0%)
DS MCS 4 - 16-QAM 3/4	225 (0.3%)	SS MCS 4 - 16-QAM 3/4	3 (0%)
DS MCS 3 - 16-QAM 1/2	93 (0.1%)	SS MCS 3 - 16-QAM 1/2	2 (0%)
DS MCS 2 - QPSK 3/4	78 (0.1%)	SS MCS 2 - QPSK 3/4	2 (0%)
DS MCS 1 - QPSK 1/2	40 (0.1%)	SS MCS 1 - QPSK 1/2	11 (0%)

i Session Drops	0 sessions
Network Entry Attempts	1
Successful Network Entries	1
Network Entry Authentication Failures	0
i Total Device Reboots	2 times
i Soft Device Reboots	1 times
i Hard Device Reboots	1 times

System: We can monitor device details such as:

- Hardware version
- Software version
- MAC Address
- Serial Number
- System Up Time

 Hardware Version	5 GHz Force 300-25 Radio (ROW/ETSI)
 Serial Number (MSN)	E6XC07KSPW9N
 Firmware Version	U-Boot IPQ40xx 2012.08.1g7be0804 (Jul 07 2021 - 15:20:55)
 Software Version	4.6.2
 Software Version (Active Bank)	4.6.2
 Software Version (Inactive Bank)	4.6.1
Device-Agent Version	2.105.48
 NTP Status	NTP Disabled
 Date and Time	13 Oct 2021, 18:03:54 GMT
 System Uptime	1 hour, 46 minutes
 Wireless MAC Address	58:c1:7a:ae:c7:7c
 Ethernet MAC Address	58:C1:7A:AE:C7:7B
Contains FCC ID(s):	Z8H89FT0017
 Read-Only Users	0
 Read-Write Users	2
 GUI User Authentication	Device Local Only
 Factory Reset Via Power Sequence	Enabled
 cnMaestro Connection Status	Connecting in 1 minute
 cnMaestro Account ID	--

Wireless: Use the Wireless Status page for key information about the radio's wireless interface and connected SM. It shows current:

- Frequency
- Channel Bandwidth
- Tx Power
- Ethernet Interface Status
- Wireless Interface Status
- Country Code

SM Details:

- Device Name
- IP Address
- Mac Address
- SM Distance
- RF session UP time
- RSSI (DL/UL)
- SNR (DL/UL)
- MCS (DL/UL)
- DL Quality
- DL Capacity
- Device Model
- Software Version

Monitor ▶ Wireless

Wireless Status	Up
Operating Frequency	5560 MHz
Operating Channel Bandwidth	20 MHz
Transmit Power	28 dBm
<i>i</i> Registered Subscriber Modules	1
<i>i</i> Ethernet Status	1 000 Mbps / Full
<i>i</i> Country	Other

Registered Subscriber Modules								
Show Details								
MAC Address	IPv4 / IPv6 Addresses	Device Name	SM Distance (km)	Session Time (hh:mm:ss)	<i>i</i> RSSI (dBm) Downlink / Uplink	SNR (dB) Downlink / Uplink	MCS Downlink / Uplink	Downlink Quality
58:C1:7A:AE:74:A8	172.42.50.219	F300-25_ae74a7	1.648	00:04:26	-41/-40	48/49	DS 9/DS 9	96 %

Throughput Chart

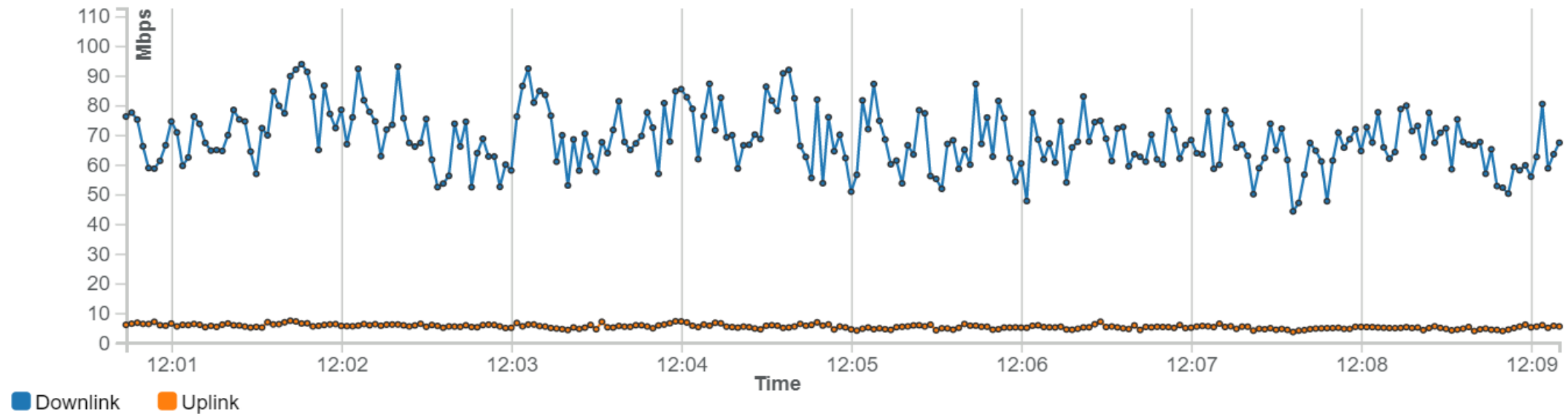
Throughput Chart: Use this page to monitor current throughput utilization in seconds, minutes or Hours.

Monitor ▶ Throughput Chart

Real-time Uplink Throughput 5.4 Mbps

Real-time Downlink Throughput 67.3 Mbps

Average Wireless Throughput



Throughput Measurement Period ☒ Seconds ☐ Minutes ☐ Hours

Network: This page displays:

- IP Details
- MTU
- Speed & Duplex
- ARP table
- Bridge table

Monitor ▶ Network

Ethernet Interface

IP Assignment	Static
 IP Address	172.42.50.218
Subnet Mask	255.255.255.248
Default Gateway	172.42.50.217
MTU Size	1 538
 Ethernet Status	1 000 Mbps / Full
Port Speed	1 000 Mbps
Port Duplex Mode	Full

ARP Table

ARP Table Show Details		
MAC Address	IP Address	Interface
04:76:B0:EA:B0:66	172.42.50.217	Bridge

Network Status

DNS Server IP	--
DHCP Option 82	Disabled
 NTP Status	NTP Disabled

Bridge Table

Bridge Table Show Details			
MAC Address	Port	Subscriber Module MAC	Aging Timer (secs)
58:C1:7A:AE:C7:7C	Own	N/A	0
58:C1:7A:AE:74:A8	WLAN	N/A	6

Syslog: Key events will be displayed here which will be helpful during troubleshooting time.

Monitor ▸ System Log

Syslog Display ☐ Disabled ☒ Enabled

Device Agent logging ☐ Disabled ☒ Enabled

Syslog File

```
Oct 3 14:17:49 F300C_226701 mgmtcmd: Error: generate credentials
Oct 3 14:01:30 F300C_226701 mgmtcmd: Error: generate credentials
Oct 3 13:45:24 F300C_226701 mgmtcmd: Error: generate credentials
Oct 3 13:28:09 F300C_226701 mgmtcmd: Error: generate credentials
Oct 3 13:14:08 F300C_226701 kernel: [ 193.019310] SM[58:c1:7a:41:f9:37] aid=1 associated with AP
Oct 3 13:12:12 F300C_226701 snmpd[2955]: DFS status: N/A
Oct 3 13:11:57 F300C_226701 snmpd[2955]: DFS status: Error!
```

Syslog File

[Download](#)

ePMP Tools



ePMP
FORCE 300

Get Cost-Effective High Speed Connectivity with the
High-gain Force 300-25 802.11AC Wave 2 solution

Software Upgrade: Use this page to update device software to take advantage of new software features and improvements.

Software Version

Radio will display two software banks, active (current) bank and inactive (backup) bank

Upgrade Options

From URL: A webserver may be used to retrieve software packages

e.g.
http://192.168.2.1/<software_upgrade_package>

From Local File: Click Browse to select the local file containing the software package.

Tools > Software Upgrade

Main Software

i Hardware Version	5 GHz Force 300-25 Radio (ROW/ETSI)
i Software Version	4.6.2
i Software Version (Active Bank)	4.6.2
i Software Version (Inactive Bank)	4.6.1
i Firmware Version	U-Boot IPQ40xx 2012.08.1g7be0804 (Jul 07 2021 - 15:20:55)
Upgrade Options	☐ URL ☒ Local File
Select File	Browse... Upgrade

Warning: Please ensure that power to the device is not interrupted during a software upgrade. Power interruption may cause flash corruption and render the device inoperable.

Backup/Restore & Factory Reset

Configuration Backup & Restore

Copy Radio Configuration to a File (Text file or encrypted/binary)

Optionally Edit the File (Text File Only)

Restore device or provision other Radios with Configuration File

Backup Configuration

Configuration File
Format

- ☒ Text (Editable)
☐ Binary (Secured)

Download

Restore Configuration

Select File

Browse...

 Skip unsupported configuration elements

☒ Disabled ☐ Enabled

Upload

Factory Default Configuration: Helps to restore factory configuration

Factory Reset Via Power Sequence:

Enabled: When Enabled, it is possible to reset the radio's configuration to factory defaults using the power cycle sequence

Disabled: When Disabled, it is not possible to factory default the radio's configuration using the power cycle sequence

Factory Default Configuration

 Reset Via Power Sequence ☐ Disabled ☒ Enabled

 Retain Passwords ☒ Disabled ☐ Enabled

 Keep Passwords ☐

 Reset to Factory Defaults

Reset

Backup Technical support File:

Technical support file can be downloaded and used for RF troubleshooting.

Backup Technical Support File

Download

Tools ▸ Spectrum Analyzer



ACS

 Channel Selection ☐ Manual ☐ DCS ☒ ACS

Rescan for the best channel

Restart ACS

Log levels *

☐ Errors

☐ Warnings

☐ Notices

☐ Info

☐ Debug

ACS DCS Log

211 20 MHz Scan List

DCS

 Channel Selection ☐ Manual ☒ DCS ☐ ACS

 SNR Difference Threshold

4

dBm | min: 3 | max: 12

 Minimal Hopping Time Interval

180

seconds | min: 90 | max: 300

 Throughput Drop Threshold

20

% | min: 10 | max: 40

Log levels *

☐ Errors

☐ Warnings

☐ Notices

☐ Info

☐ Debug

ACS DCS Log

211 20 MHz Scan List

eAlign: Use this page to monitor real time RSSI value at the time of antenna alignment

 Operating Frequency 5560 MHz

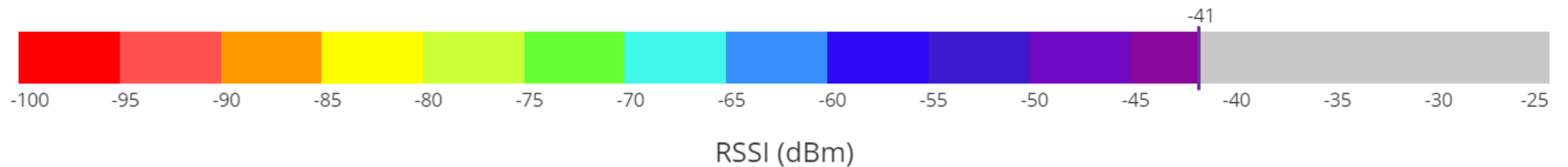
Registered SM MAC Address 58:C1:7A:AE:74:A8

Peak RSSI -41 dBm

Current RSSI -41 dBm
(Ch0: -50 dBm, Ch1: -42 dBm)

Reset Measurements

Reset



Wireless link test: Use this page to conduct a simple test of wireless throughput. This allows user to determine the throughput that can be expected on a particular link without having to use external tools.

Test Setup

SM MAC Address

58:C1:7A:AE:74:A8

Packet Size

☐ Small (128 bytes)

☐ Medium (800 bytes)

☒ Large (1500 bytes)

Duration

☒ 4 seconds

☐ 10 seconds

☐ 20 seconds

Start Test

Downlink

--

Uplink

--

Aggregate

--

Registered Subscriber Modules

Show Details

MAC Address	IPv4 / IPv6 Addresses	Device Name	SM Distance (km)	Session Time (hh:mm:ss)	RSSI (dBm) Downlink / Uplink	SNR (dB) Downlink / Uplink	MCS Downlink / Uplink
58:C1:7A:AE:74:A8	172.42.50.219	F300-25_ae74a7	1.648	3 days 18:20:57	-41/-41	48/48	DS 7/DS 9

Watchdog:

Watchdog performs ping checks every several minutes to determine the reachability of a target IP address. If IP address is unreachable, a chosen action is performed.

Watchdog Action

Wireless Restart: In case of lost ping connectivity to the specified IP address, the device automatically restarts the wireless interface.

Ethernet Restart: In case of lost ping connectivity to the specified IP address, the device automatically restarts the Ethernet interface.

Device Reboot: In case of lost ping connectivity to the specified IP address, the device automatically reboots.

Watchdog

i Watchdog

☐ Disabled ☒ Enabled

i Watchdog Action

☒ Wireless Restart ☐ Ethernet Restart ☐ Device Reboot

i IP Address

i Watchdog Ping Interval

minutes | min: 1 | max: 60

i Watchdog Ping Retries

min: 3 | max: 30

Ping

IP Address Version

☒ IPv4 ☐ IPv6

IPv4 Address, Fully Qualified Domain Name, or Hostname

10.31.255.25

Number of Packets (-c)

10

min: 1 | max: 100

Buffer Size (-s)

min: 1 | max: 65507

TTL (-t)

min: 1 | max: 128

Start Ping

Ping Results

Traceroute

IPv4/IPv6 Address, Fully Qualified Domain Name, or Hostname

10.31.255.25

Fragmentation (-F)

☒ OFF ☐ ON

Trace method (-I)

☒ ICMP ECHO ☐ UDP

Display TTL (-l)

☒ OFF ☐ ON

Verbose (-v)

☒ OFF ☐ ON

Start Traceroute

Traceroute Results

ePMP Troubleshooting



ePMP
FORCE 300

Get Cost-Effective High Speed Connectivity with the
High-gain Force 300-25 802.11AC Wave 2 solution

When a problem occurs, attempt to answer the following questions:

What is the history of the problem?

- Have we changed something recently?
- Have we seen other symptoms before this?

How wide-spread is the symptom?

- Is the problem on only a single SM? (If so, focus on that SM.)
- Is the problem on multiple SM's? If so
 - Is the problem on one AP in the cluster? (If so, focus on that AP)
 - Is the problem on multiple, but not all, aps in the cluster? (If so, focus on those APs)
- Is the problem on all APs in the cluster? (If so, focus on the site infrastructure)

Based on data in the System Log

- is intermittent connectivity indicated? (If so, verify your configuration, power level, SNR, cables and connections, and the speed duplex of both ends of the link).
- are connections made via *shielded* cables?

S. N.	Issues	Resolution Steps
1	Interference	1. Do the Wireless Link Test in BTS/AP and See if UL throughput is not as per requirement. 2. RUN Spectrum Analyzer and find least noise level frequency. 3. Configure frequency as per Spectrum result 4. RSSI should be as per Link Budget. 5. CINR (RSSI and Noise level difference) should be ≥ 10
2	Packet drops in CPE IP	
3	Latency in CPE Management IP	
4	CPE is not Latching	In CPE/SM: 1. Goto Monitor---> Wireless and see if any error message under Registration Status column and If yes or No under Meeting criteria column. 2. Check if added correct SSID, Security and Preshared Key in Preferred AP list. 3. NetworkEntryRSSIThreshold should be -90 In BTS/AP: 1. in SSID, Channel BW, Frequency Channel, Security and WPA2 preshared key are same as CPE/SM. 2. CPE/SM distance to AP Ok? Refer Configuration--Radio--Range 3. Check SM Registration allowed count is $\geq$ No. of CPE devices 4. SM Receive Level should be -40
5	Packet drops in WAN IP	1. Check Ethernet Speed & Duplex configuration & Status in Mux/RiCi/Switch, BTS UBR, CPE UBR and CE Router. 2. Check if the RSSI level as per Link Budget. 3. Check if the Interference on operating frequency. 4. Change the least noise level frequency as per spectrum analyzer result. 5. CINR (RSSI and Noise level difference) should be ≥ 10
6	Duplex Mismatch	1. Check the Ethernet Speed & Duplex Configuration in BTS & Customer UBR (By default, It should be AUTO). 2. Check Ethernet Speed & Duplex configuration & Status in Mux/RiCi/Switch and CE Router. 3. Change Ethernet Speed & Duplex in BTS & Customer UBR as per the configuration in Mux/RiCi/Switch and CE Router. For example, if CE Router is configured as Manual/Forced 100Mbps/Full, then configure the same in both UBRs.

S.N.	Issue Statement	Resolution Steps
8	Device going into default Mode	1. For BTS end - ePMP1000 radios(new supply), Upgrade the radio with 4.5.6 Software version, change TDD ratio as 50/50 & configure synchronization source as GPS.If GPS configuration is not available ,follow below step mentioned in point 2. 2. For BTS end - without GPS Sync ePMP1000 radios(old supply), check DC input power level for any Fluctuation, upgrade the device with latest software, & disable "Reset via Power Sequence" option. For repeated cases, change admin password of the device (password format to be decided & maintained by Airtel circle team). 3. For Customer end - check AC input power level for any Fluctuation, upgrade the device with latest software, & disable "Reset via Power Sequence" option. For repeated cases, change admin password of the device (password format to be decided & maintained by Airtel circle team).
9	Antenna alignment done but RSSI is not improving	1. Check the key configuration parameters like TX power, antenna gain & target receive level. These parameters have direct impact on RSSI level & if configured incorrect, will always result in to wrong RSSi value. 2. Even after correct configuration, if the RSSI level is not decreasing, then take the LAT/LONG of both ends & check LOS in LINK PLANNER.
10	Device IP is not pinging	1. Check whether the LAN port of radio is coming UP or not. 2. Bring the radio down & check with Small cable. If the ping is OK with small cable then check the cat6 cable between POE & Radio and recipm the RJ45 connector at both end. If ping is not OK with Small cable also, in that cases arrange spare POE & Radio to check.
11	LAN Port is Down	1. Bring the radio down & check with Small cable. If the ping is OK with small cable then check the cat6 cable between POE & Radio and recipm the RJ45 connector at both end. If ping is not OK with Small cable also, in that cases arrange spare POE & Radio to check.

S.N.	Issues Statement	Resolution Steps
12	RF Throughput Issue	1. Check whether the correct channel BW is configured in radio as per throughput requirement. 2. Check whether the RSSI level is as per recommendation for a given distance. 3. If above Parameters are OK, then check for interference at both BTS & customer end, select the frequency having least noise as per spectrum analyser result. 4. If there is very heavy interference & no frequency is clean enough to give desired throughput, in that case try to relocate the antenna position on tower to avoid interference from any nearby device.
13	Bandwidth (Speed Test) Issue	1. First step is to check the BW at BTS MUX/Switch port. 2. If above step is OK, Check the RF throughput as described in "RF throughput Issue". 3. In case RF throughput is OK, check the BW at customer end & if the issue persists, check the Ethernet speed & duplex configuration in BTS Mux/Switch/RiCi, CE router/switch & UBRs.
14	WAN IP ping Ok with Laptop but not Ok with CE RTR	1. Check Duplex Mismatch in UBR and CE RTR 2. Shut/Unshut/Reboot CE RTR and check 3. Change RTR port or Replace the RTR and check.
15	Device is not logging	1. Verify Management VLAN and Management IPs or Backdoor IP configured in Laptop. 2. Ping device Management IP/ Backdoor IP (169.254.1.1) 3. Clear the browser history and restart the browser. 4. Check with other laptop. 5. Contact OEM.
16	BGP flapping	1. Check RF session time. 2. Do the Wireless Link Test in BTS/AP and See if UL throughput is not as per requirement. 3. RUN Spectrum Analyzer and find least noise level frequency. 4. Configure frequency as per Spectrum result 5. RSSI should be as per Link Budget. 6. CINR (RSSI and Noise level difference) should be ≥ 10

SNo.	Issue Statement	Resolution Steps
18	CE Router Port is not coming UP	1. Check Duplex setting in UBR and CE RTR. 2. Shut/Unshut CE RTR Port. 3. Change CE RTR Port and check. 4. Change LAN cable between POE and CE RTR.
19	Device is login on NMS	1. Change the browser and check. 2. SSH device and Reboot the device and check. 3. Check device IP ping on load of 1472.
20	Spectrum Analyzer not running	1. Check JAVA available in laptop. JAVA latest version should be in laptop. 2. Change device Mode as Spectrum Analyzer (Configuration -->Radio--Spectrum Analyzer). 3. Reboot the device and relogin. 4. Goto Tools--> Download Spectrum Analyzer 5. Add device IP in Java exception list 6. Run Spectrum Analyzer downloaded files and Check.
21	Firmware upgradation	1. FE should have latest software available with him. If not, one can download the same from support.cambiumnetworks.com/downloads. 2. Once you have the required software, login in to device, go to TOOLS --> SOFTWARE UPGRADE, browse the software from your laptop & upgrade. After the upgarde is complete, reboot the device.
22	Customer application is not working	1. Check ARP table in PE RTR and verify CE RTR IP/MAC. 2. Check CPE UBR firmware versions. 3. Change the FW and check. 4. Replace the CPE UBR and check.
23	Multicast is not working	1. Enable Multicast configuration in BTS and CPE UBR. 2. Contact OEM.
24	Few sites are not browsing	1. Check backend provisioning.
25	Frequently link down	1. Check RSSI as per link budget. 2. Check Interference on operating frequency. 3. Change the frequency as per Spectrum Analyzer result.

Monitoring & Optimizing Retransmission Packets in order to minimize packet Drops on Small Cell(Mobility) Links.

1. Where to check Retransmission Packet Percentage & acceptable value ?

We can check Retransmission packet percentage in **Monitor → Performance → Wireless Statistics** and for smooth link performance retransmission should be **less than or equal to 2%**

2. How to minimize Retransmission Packet percentage ?

Step 1: Run spectrum analyzer and select the best available clean channel & configure the same in Access point.

Step 2: Run Wireless link test 8-10 times for test duration of 20 Seconds & Check for Retransmission.

If Retransmission reduced to 2% or below, then you are ready to deliver the link but if not, then follow below steps:

Step 3: Go to **Monitor → Performance** and check the packet percentage against each MCS in both UL & DL

Step 4: Find out the Minimum MCS value having significant packet percentage in both UL & DL

Step 5: Fix the DL MCS in AP and UL MCS in SM as per observation of Step 4.

Step 6: Run Wireless link test 8-10 times for test duration of 20 Seconds & Check for Retransmission.

If you have selected and fixed to correct MCS value, then retransmission will definitely reduce.

During device configuration or after completing the configuration, one should always re-verify below parameters, for smoother link delivery.

1. **RSSI:** Being a key parameter, RSSI should be always as per recommendation.(Refer Distance Vs Throughput Chart).
2. **Frequency:** Once the desired RSSI is achieved, one should select frequency as per spectrum analyzer result. This will reduce outage caused by interference. (Refer Guide to use Spectrum analyzer)
3. **EIRP:** EIRP is sum of transmit power & antenna gain & it should be 36 in case of P2MP link.
(EIRP=TX Power + Antenna Gain- Cable Loss). In our case, BTS antenna gain is 15dBi so TX power should be configured as 21 & at customer end TX power should be auto.
4. **Subscriber Target Receive level:** This parameter is available BTS radio configuration & it should be configured as -40. Lets say, for example we configure this value as -60, then in that case Uplink RSSI value will not go below -60dBm.
5. **Network entry threshold:** This parameter is available in Customer radio configuration & it should be configured as -90. Lets say, for example we configure this value as -60, then in that case link will get registered only if RSSI value on the link is below -60dBm.
6. **Port Speed & Duplex Configuration:** By default it is configured as auto, but FE should must confirm the customer router/switch port speed & duplex configuration & configure the radios accordingly.
7. **Frequency Selection at customer Radio:** One should select all frequencies in all channel bandwidth in customer radio. It makes operations task easier in case of interference.
8. **Wireless Pre-shared Key:** Should be same at the both ends. If there is any mismatch link will not get registered.
9. **SM traffic Isolation:** By default it is disabled, but it should always be enabled to prevent unwanted traffic from one CPE to another.
10. **Management Access:** This parameter is available in BTS radio configuration & should be selected as Ethernet & Wireless or else one would not be able to login the BTS radio from customer end.
11. **Reset through Power Sequence:** It should be disabled at the time of radio configuration in order to avoid auto factory reset in case of power fluctuation.
12. **Username & Password:** Device username & password should be changed in order to avoid any unwanted login. Password Format to be decided by Airtel.

Do's

- Each AP should have unique SSID and device name.
- Always add 'Preferred AP' on Subscriber Modules
- Use always Cambium supplied PoE along with equipment.
- Keep 'Max Range' value in AP at least One or two km above the distance of Subscriber Module.
- Always enable 'SM traffic isolation'
- Use 'eAlign' tool to align and get good RSSI on Subscriber Modules
- Use 'Wireless Link test' to check capacity of link during installation and operation.

Don't

- Do not leave SA in Enable mode once you choose a free channel.
- Do not keep passwords in default settings.
- Do not frequently remove or connect RJ-45 cable in POE+DATA port of POE.
- Do not allow ethernet cable length >100m

cnMaestro Overview



Course Contents

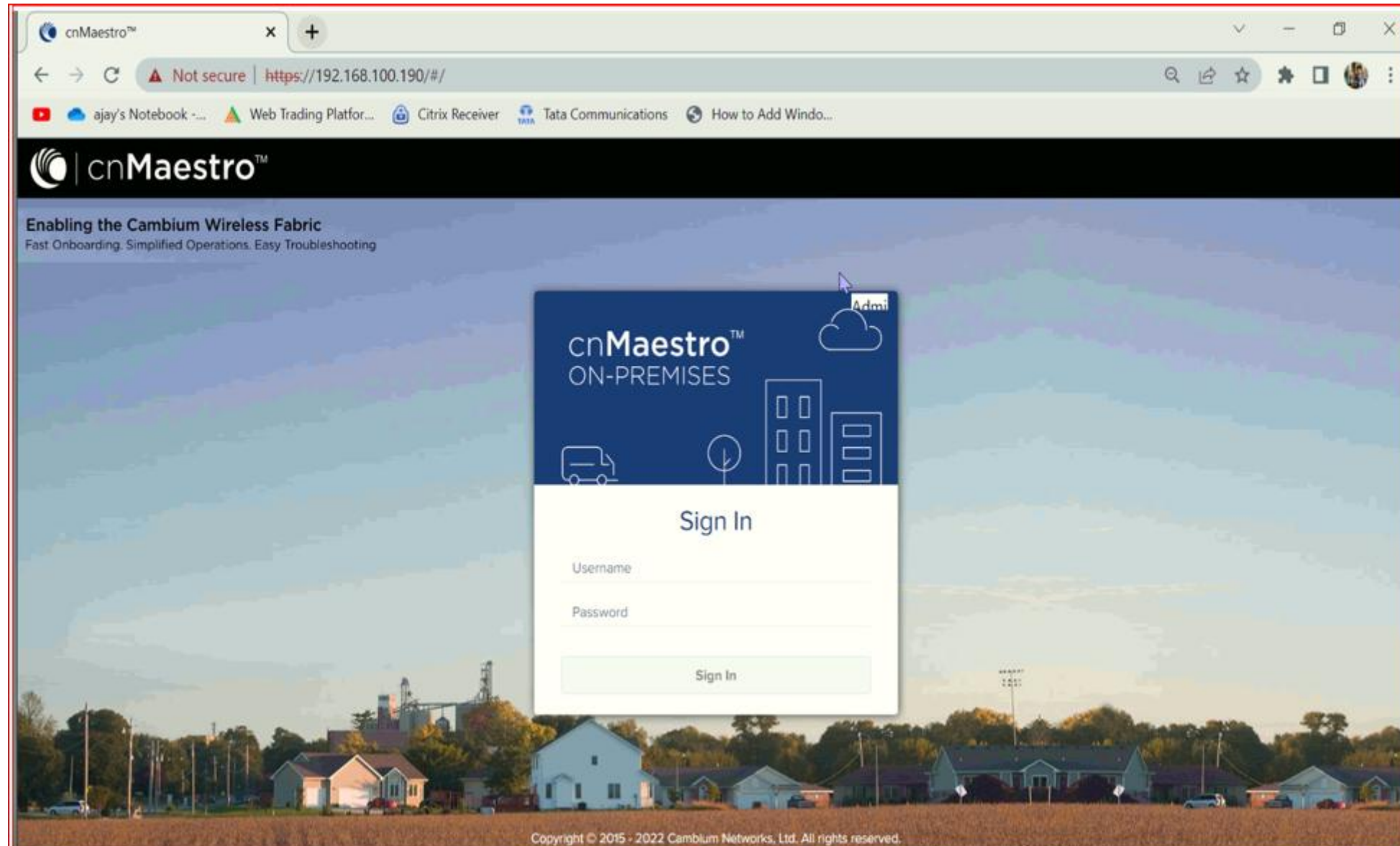
- ❑ Overview, Features and Functionality of cnMaestro
- ❑ Login Process
- ❑ Creating Tower Groups
- ❑ Create Cambium ID and Onboarding Key
- ❑ Creating Users
- ❑ Device Onboarding Process
- ❑ What can be done before device Approve to Onboard?
- ❑ Reports (Device Status, Performance, Active Alarms, Alarms History and Events)
- ❑ Inventory and Statistics Report Export
- ❑ Single & In Bulk devices firmware upgrade
- ❑ Creating configuration Templates and Apply on Single & In Bulk Devices configuration
- ❑ Single & In Bulk Devices configuration Backup
- ❑ Server backup
- ❑ Monitor Server Health
- ❑ Alarms forward on mail based on Severity
- ❑ Audit Logs
- ❑ Syslog
- ❑ Technical Support Dump

cnMaestro - End-to-End Network Control

cnMaestro is a cloud-based or on-premises software platform for secure, end-to-end network control. The program's wireless network manager simplifies device management by offering full network visibility and zero touch provisioning. View and perform a full suite of wireless network management functions in real time. Optimize system availability, maximize throughput, and meet emerging needs of business and residential customers.

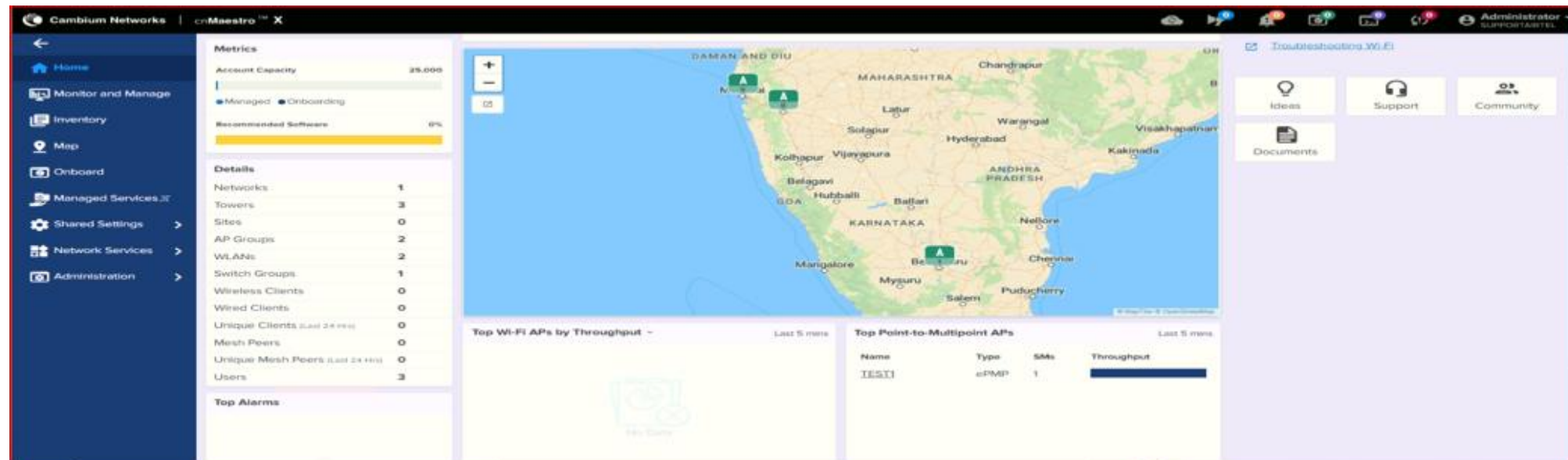
Login CnMaestro NMS

- ❖ CnMaestro IP
10.55.88.142
- ❖ Enter User Name and
Password



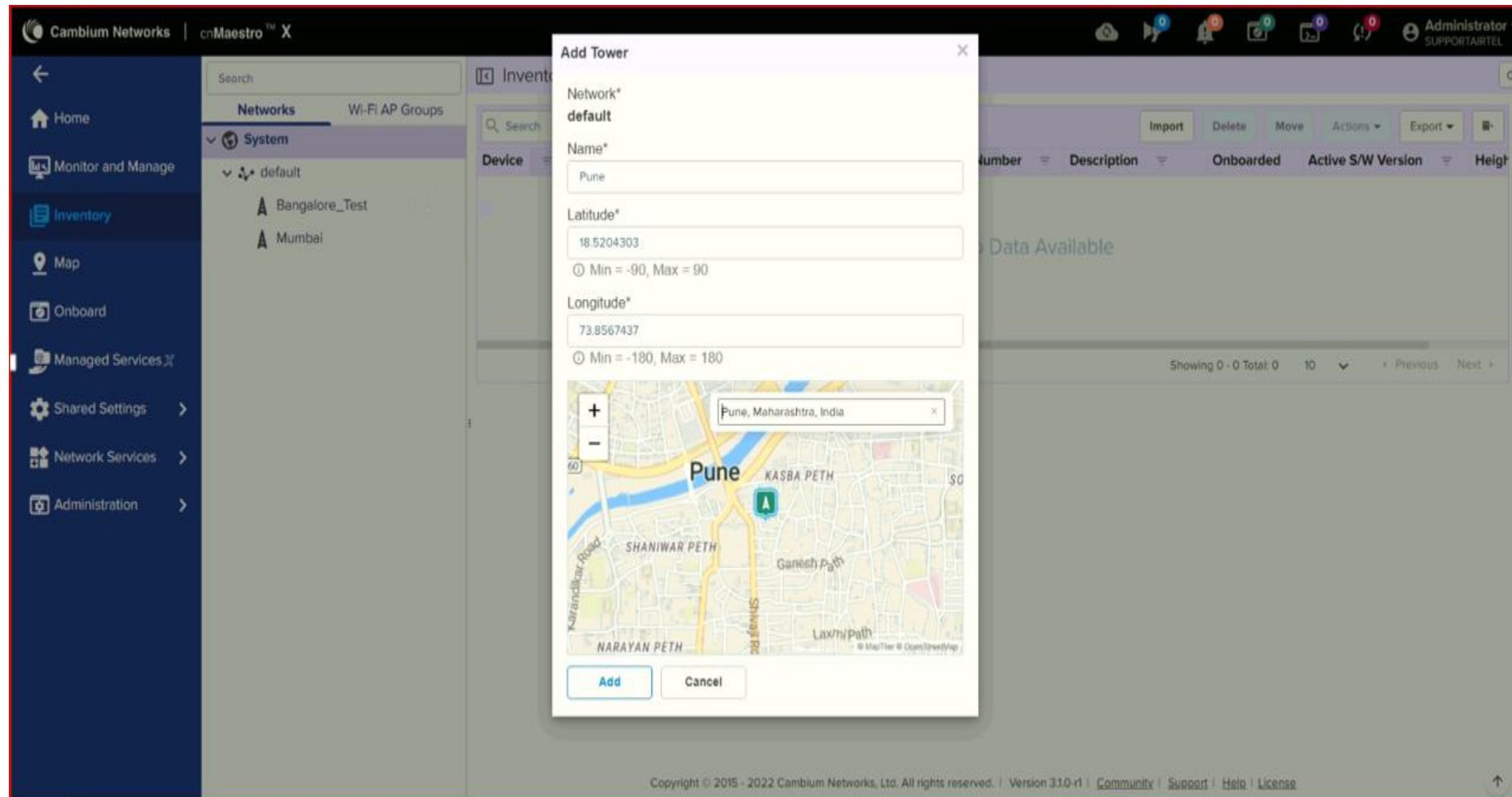
CnMaestro Dashboard

- ❖ Total devices
- ❖ Offline Devices
- ❖ Onboarding device
- ❖ Connection health
- ❖ Alarms (Critical, Major and Minor) in 24Hrs
- ❖ No of Networks, Towers and Users
- ❖ Account Capacity



Creating Tower Group

- ❖ Configure Tower Name
- ❖ Find Tower name in Map and locate for Latitude and Longitude



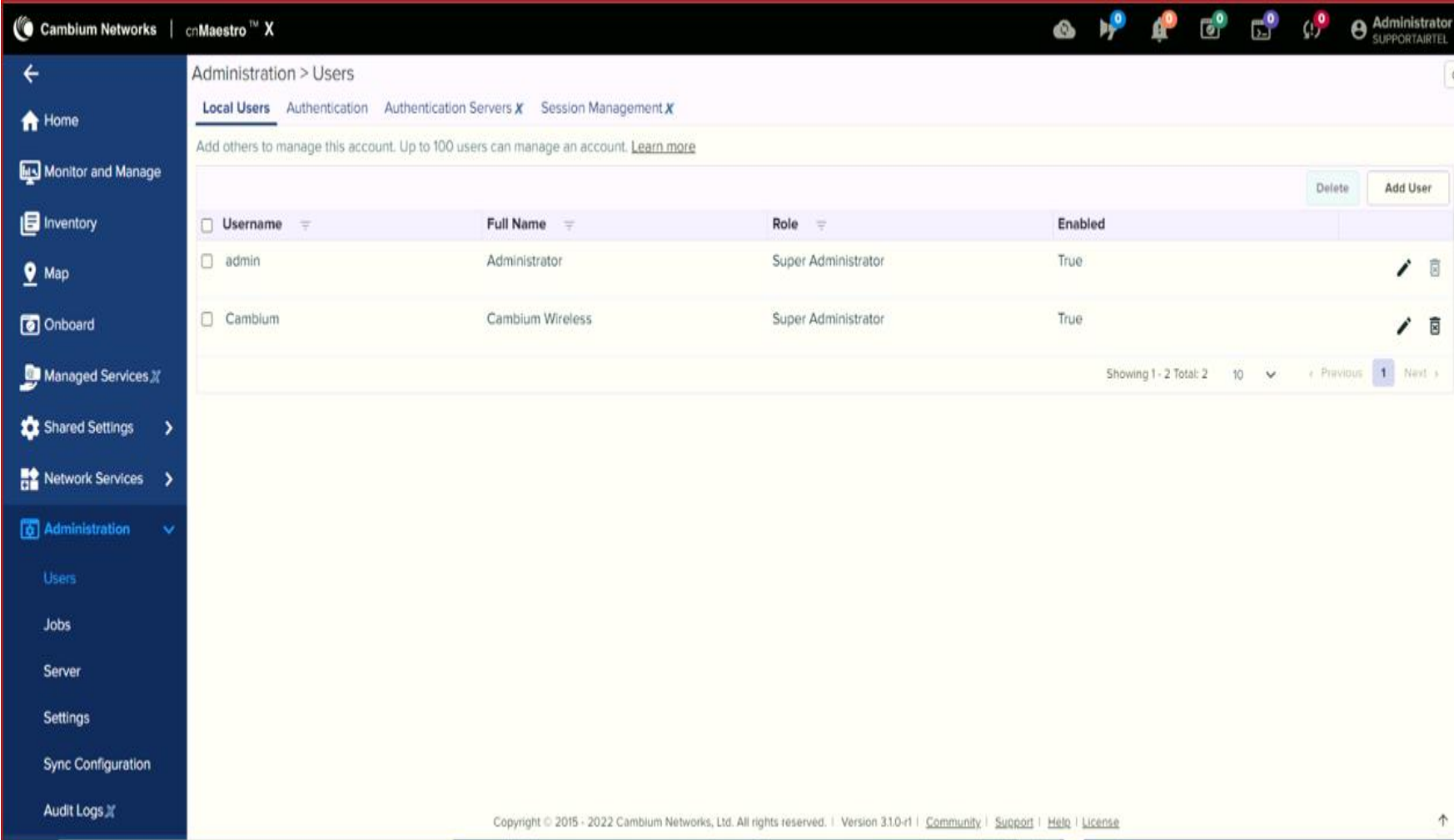
Creating Cambium ID and Onboarding key

Each user can have their own Onboard Key
Up to 100 users can manage an account.

Steps:

1. Create a User (User Name, Full Name, Select User Role),
2. Create Onboard Key for created user.

Creating Users



Administration > Users

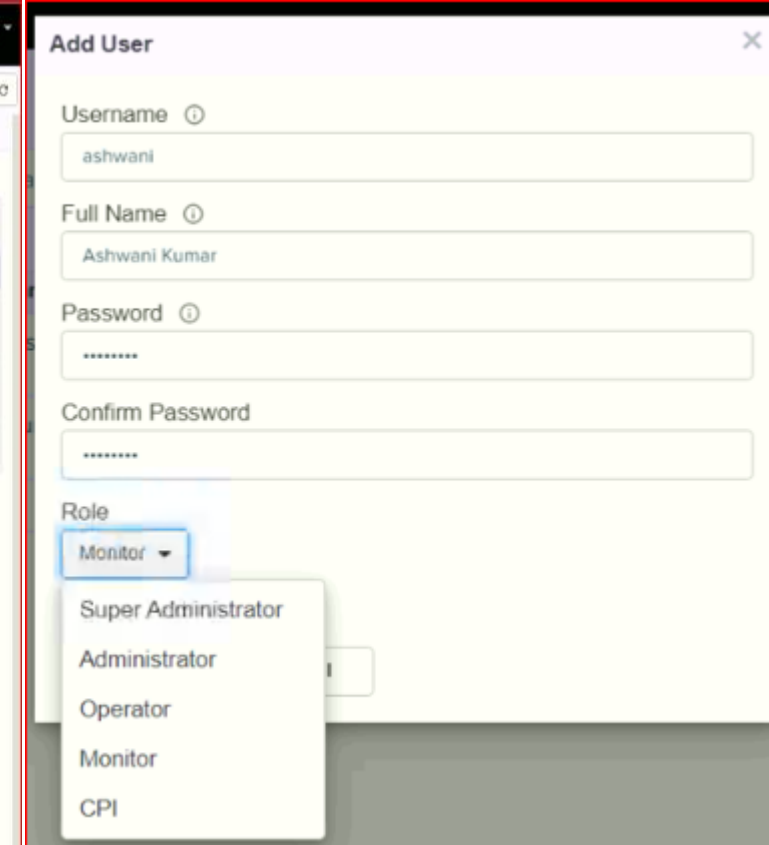
Local Users Authentication Authentication Servers X Session Management X

Add others to manage this account. Up to 100 users can manage an account. [Learn more](#)

☐	Username	Full Name	Role	Enabled	
☐	admin	Administrator	Super Administrator	True	 
☐	Cambium	Cambium Wireless	Super Administrator	True	 

Showing 1 - 2 Total: 2 10 < Previous 1 Next >

Copyright © 2015 - 2022 Cambium Networks, Ltd. All rights reserved. | Version 3.1.0-r1 | [Community](#) | [Support](#) | [Help](#) | [License](#)



Add User

Username 
ashwani

Full Name 
Ashwani Kumar

Password 

Confirm Password

Role
Monitor 
Super Administrator
Administrator
Operator
Monitor
CPI

- Super Administrator – Super Administrators can perform all operations.
- Administrator – Administrators can modify cnMaestro application functionality, but they are not able to edit User, API, or Server configuration.
- Operator – Operators are able to configure device-specific parameters and view all configuration.
- Monitor - Monitors have only view access.
- CPI - CPI can perform onboarding the devices using the CBRS tool and has the view access only

Creating Onboarding Key

 Cambium Networks

cnMaestro™ X

 Administrator
SUPPORTAIRTEL

←

Home

Monitor and Manage

Inventory

Map

Onboard

Managed Services

Shared Settings

Network Services

Administration

Onboard

Devices 60 GHz cnWave Network Settings

Cambium ID: SUPPORTAIRTEL

☒ Enable Cambium ID based authentication to onboard devices

Enabling this feature allows a device to be claimed by entering the Cambium ID and Onboarding Key on the device. This information can be set on the device via its user interface (or SNMP or CLI on some devices). Each user can have their own Onboarding Key. [Learn more](#)

The following users can claim devices using the cnMaestro Cambium ID and the user's Onboarding Key.

User:	Administrator	Onboarding Key:			Delete
User:	Cambium Wireless	Onboarding Key:			Delete

Save

Cancel

Add New

Copyright © 2015 - 2022 Cambium Networks, Ltd. All rights reserved. | Version 3.1.0-r1 | [Community](#) | [Support](#) | [Help](#) | [License](#)

↑

Device Onboarding Process

Device can be On board through two simple way.

Device configuration::

1. Configure the CnMaestro configuration in device and save.
2. Device will display onboard section and
3. Approve/Unapprove on NMS.

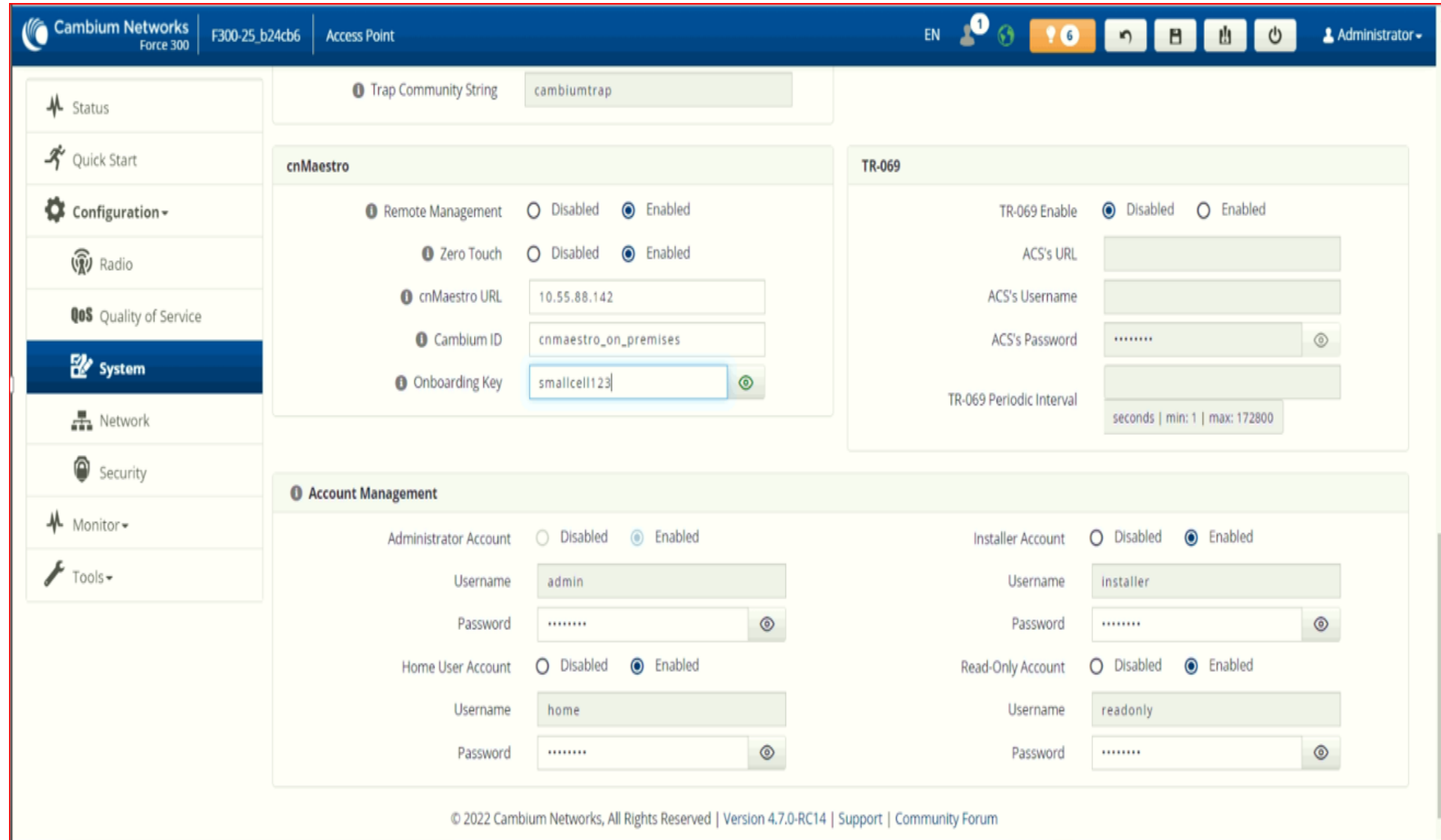
Claim the device on NMS

- Claim the device with device ESN number on NMS and Approve first
- Single or Multiple devices can be claimed
- Enter devices ESN number with comma-separated or one ESN per Line
- Device will be Onboard automatically once reachable to NMS

Preferred device configuration to Onboard the device as ESN number is to be known to claim the device.

Device configuration to On board on NMS

- ✓ CnMaestro URL
 - ✓ Cambium ID
 - ✓ Onboarding Key
-
- ✓ cnMaestro URL: 10.55.88.142
 - ✓ Cambium ID: cnmaestro_on_premises
 - ✓ Onboarding key: smallcell123



Cambium Networks Force 300 F300-25_b24cb6 Access Point

EN 1 6 Administrator

Trap Community String cambiumtrap

cnMaestro

Remote Management ☐ Disabled ☒ Enabled

Zero Touch ☐ Disabled ☒ Enabled

cnMaestro URL 10.55.88.142

Cambium ID cnmaestro_on_premises

Onboarding Key smallcell123

TR-069

TR-069 Enable ☒ Disabled ☐ Enabled

ACS's URL

ACS's Username

ACS's Password

TR-069 Periodic Interval seconds | min: 1 | max: 172800

Account Management

Administrator Account ☐ Disabled ☒ Enabled

Username admin

Password

Home User Account ☐ Disabled ☒ Enabled

Username home

Password

Installer Account ☐ Disabled ☒ Enabled

Username installer

Password

Read-Only Account ☐ Disabled ☒ Enabled

Username readonly

Password

© 2022 Cambium Networks, All Rights Reserved | Version 4.7.0-RC14 | Support | Community Forum

Device waiting for Approval

Cambium Networks

cnMaestro™ X

←

Home

Monitor and Manage

Inventory

Map

Onboard

Managed Services

Shared Settings

Network Services

Administration

Onboard

Devices60 GHz cnWave NetworkSettings

The Onboarding Queue holds devices before they are added to the network. They are approved by setting location, configuration, or software. You can pre-provision devices before the onboarding process and be managed by cnMaestro. You can pre-provision devices before the onboarding process and be managed by cnMaestro. You can pre-provision devices before the onboarding process and be managed by cnMaestro.

Search

Type	Serial Number	Device
ePMP Force 300-2...	E6XG0CWLRTWW	F300-25
ePMP Force 300-2...	E6YB048J6ZLX	F300-25

Note: Devices will remain in the queue for 1 week after onboarding.

Edit Device Configuration: F300-25_b24cb6

Details

Location

Update

Configure

Configure Device Location

Network

default

Tower

None

Search

None

Bangalore_Test

Mumbai

Pune

Map

Save

Cancel

Claim Device

Approve All

Export

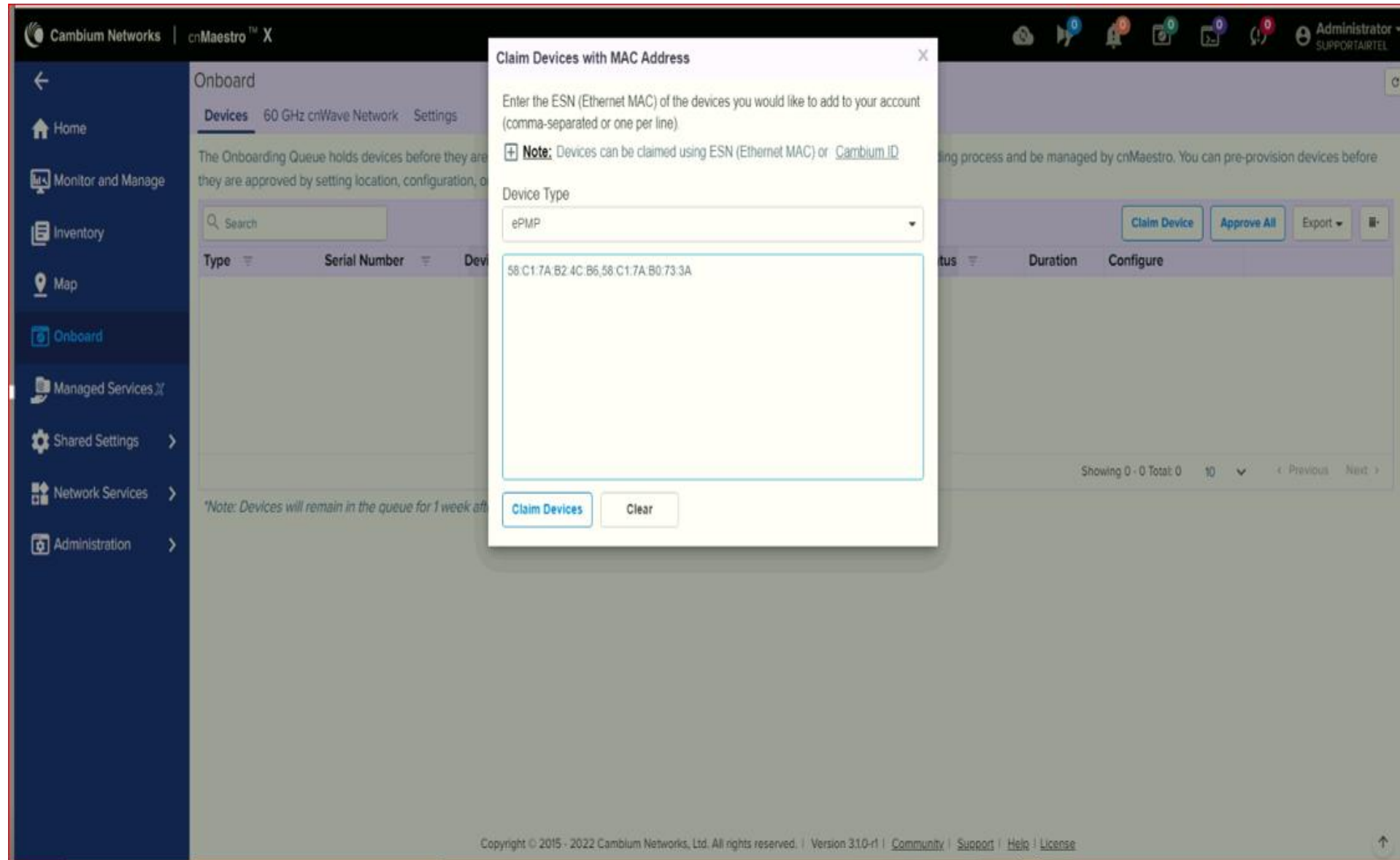
Status	Duration	Configure
Waiting for A...	0d 0h 3m	ApproveDelete
Waiting for A...	0d 0h 3m	ApproveDelete

Showing 1 - 2 Total: 210Previous1Next

Copyright © 2015 - 2022 Cambium Networks, Ltd. All rights reserved. | Version 3.1.0-r1 | Community | Support | Help | License

Claim device on NMS

- Device configuration not required
- ESN Number required to claim on NMS
- Enter ESN Number and Approve



Claim device on NMS

 Cambium Networks

cnMaestro™ X

 Home

 Monitor and Manage

 Inventory

 Map

 Onboard

 Managed Services

 Shared Settings

 Network Services

 Administration

Onboard

Devices60 GHz cnWave NetworkSettings

The Onboarding Queue holds devices before they are added to your account. Devices must be approved in order to complete the onboarding process and be managed by cnMaestro. You can pre-provision devices before they are approved by setting location, configuration, or software version. [Learn more](#)

Search

Claim DeviceApprove AllExport

Type	Serial Number	Device	MAC	IP Address	Added By	Status	Duration	Configure
ePMP	N/A	ePMP-B0733A	58:C1:7A:B0:73:3A	N/A	Administrator Using MAC Address	Waiting for D...	< 1m	ApproveDelete
ePMP	N/A	ePMP-B24CB6	58:C1:7A:B2:4C:B6	N/A	Administrator Using MAC Address	Waiting for D...	< 1m	ApproveDelete

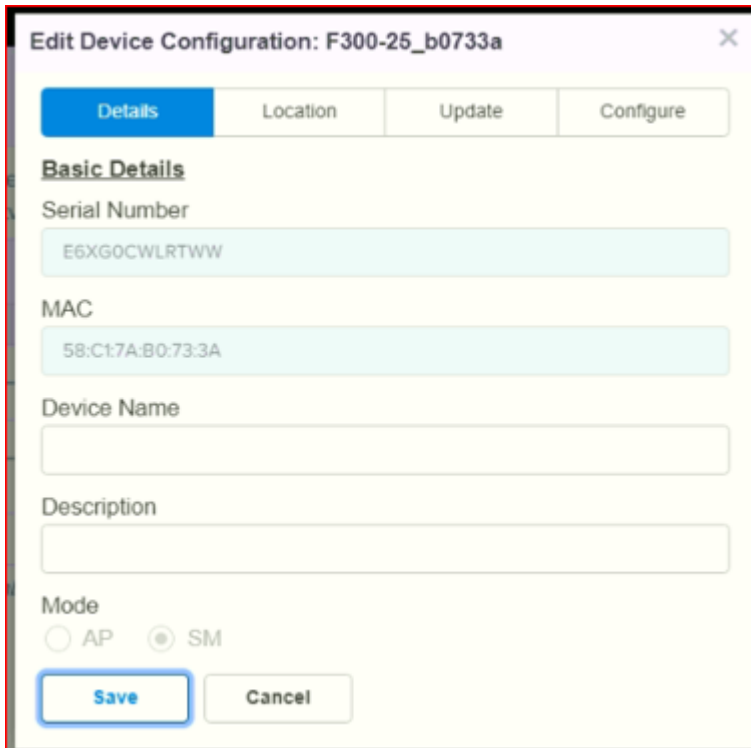
Showing 1 - 2 Total: 210Previous1Next

*Note: Devices will remain in the queue for 1 week after onboarding successfully.

Copyright © 2015 - 2022 Cambium Networks, Ltd. All rights reserved. | Version 3.1.0-r1 | [Community](#) | [Support](#) | [Help](#) | [License](#)

What can be done before Approve?

- Basic configuration details like Device Name, description
- Tower group allocation
- Software upgrade
- Configuration with pre created templates



Edit Device Configuration: F300-25_b0733a

Details Location Update Configure

Basic Details

Serial Number
E6XG0CWLRTWW

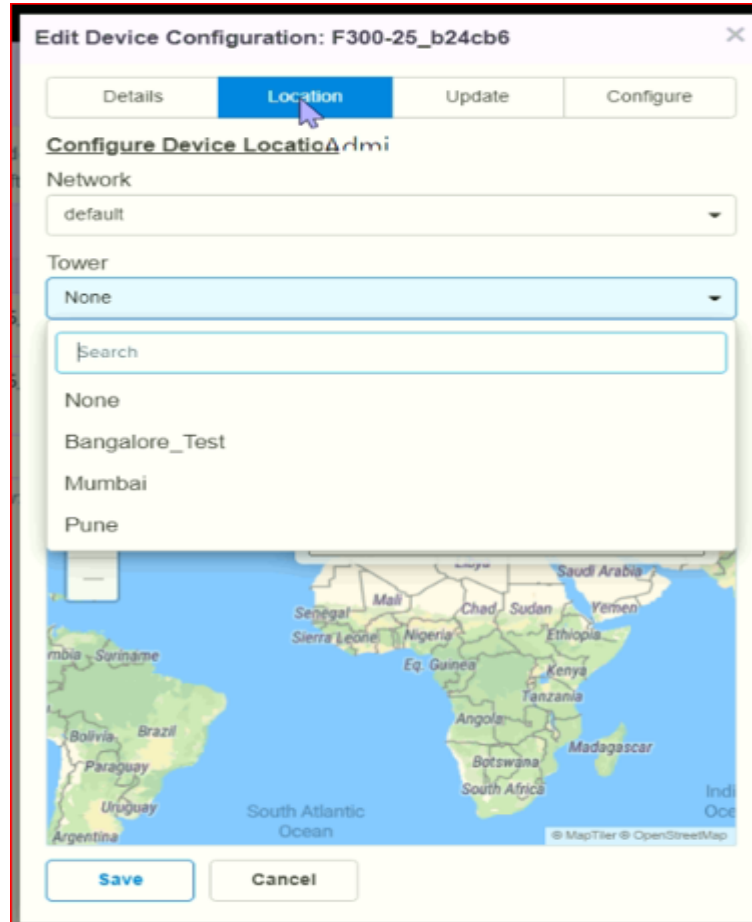
MAC
58:C1:7A:B0:73:3A

Device Name

Description

Mode
☐ AP ☒ SM

Save Cancel



Edit Device Configuration: F300-25_b24cb6

Details Location Update Configure

Configure Device Location

Network
default

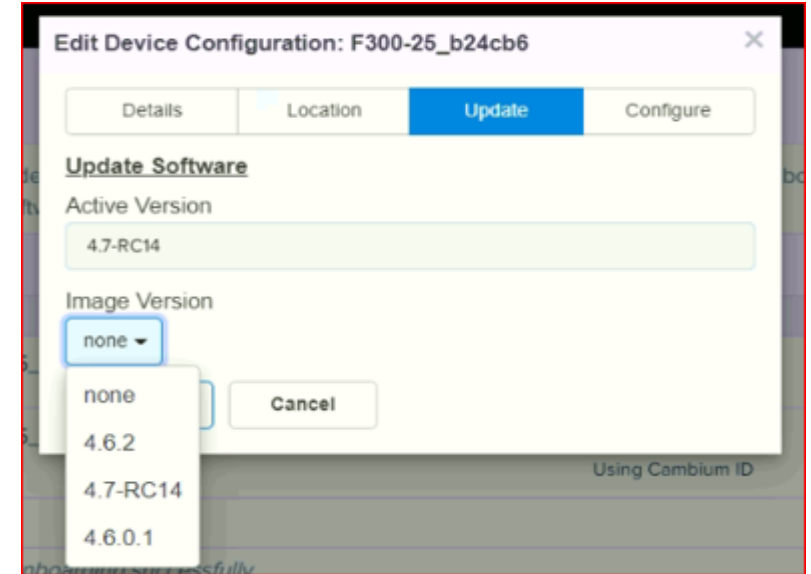
Tower
None

Search

None
Bangalore_Test
Mumbai
Pune



Save Cancel



Edit Device Configuration: F300-25_b24cb6

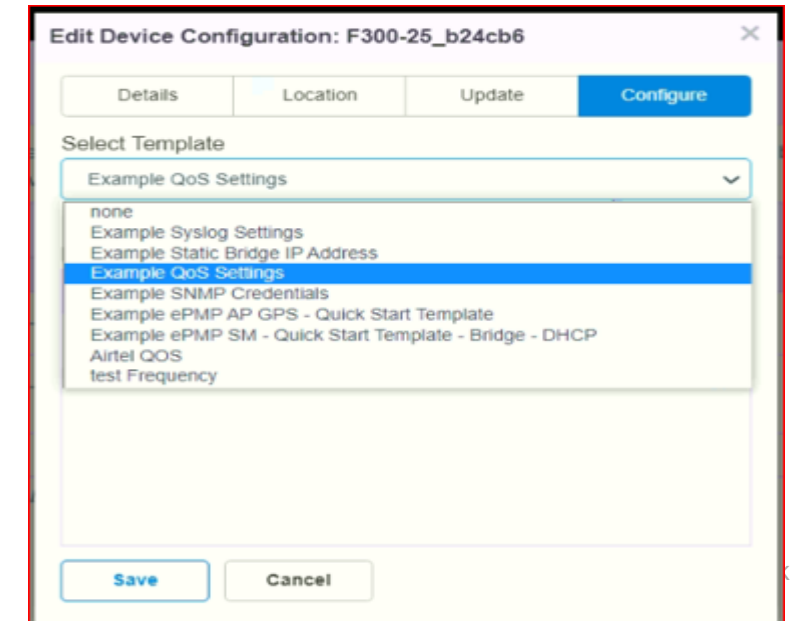
Details Location Update Configure

Update Software

Active Version
4.7-RC14

Image Version
none

Cancel



Edit Device Configuration: F300-25_b24cb6

Details Location Update Configure

Select Template

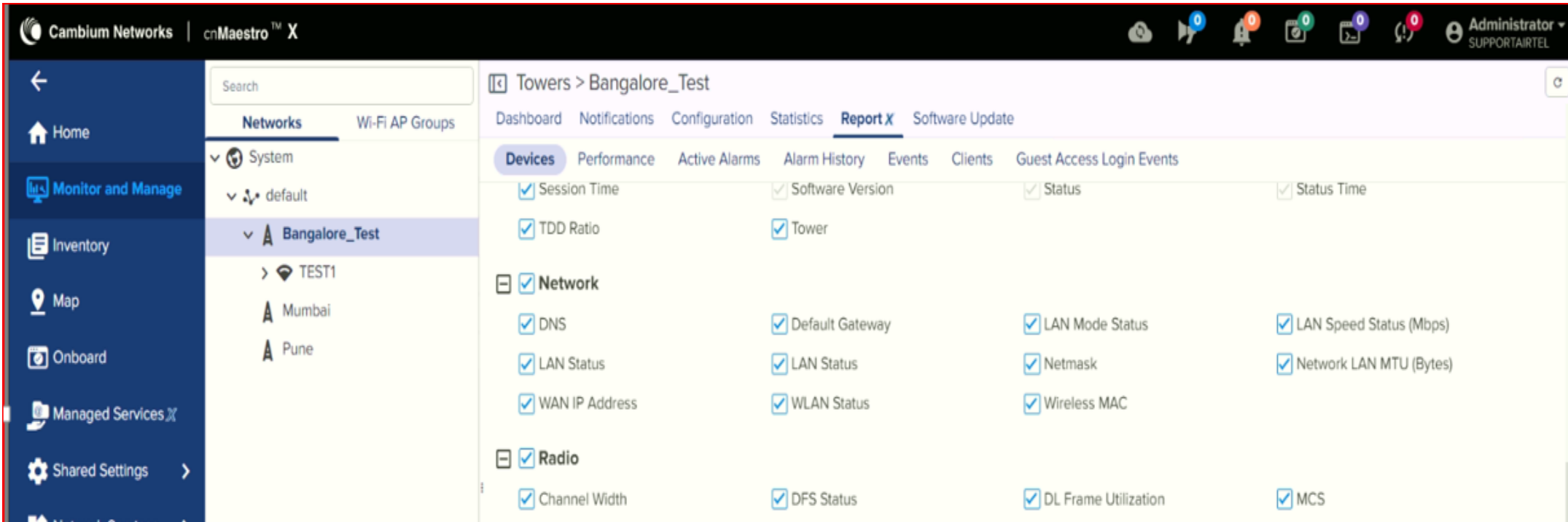
Example QoS Settings

none
Example Syslog Settings
Example Static Bridge IP Address
Example QoS Settings
Example SNMP Credentials
Example ePMP AP GPS - Quick Start Template
Example ePMP SM - Quick Start Template - Bridge - DHCP
Airtel QOS
test Frequency

Save Cancel

Reports

- ❖ Device Status,
- ❖ Performance,
- ❖ Active Alarms,
- ❖ Alarms History
- ❖ Events



The screenshot displays the Cambium Networks cnMaestro X web interface. The top navigation bar includes the Cambium Networks logo, the product name 'cnMaestro™ X', and a user profile for 'Administrator' with a 'SUPPORTAIRTEL' link. A sidebar on the left contains navigation icons for Home, Monitor and Manage, Inventory, Map, Onboard, Managed Services, and Shared Settings. The main content area is titled 'Towers > Bangalore_Test' and features a breadcrumb trail: Dashboard > Notifications > Configuration > Statistics > Report X > Software Update. Below this, a sub-navigation bar includes 'Devices' (selected), Performance, Active Alarms, Alarm History, Events, Clients, Guest Access Login, and Events. The 'Devices' section is expanded, showing a list of metrics with checkboxes for selection. The metrics are organized into three main categories: 'Session Time', 'TDD Ratio', and 'Network'. The 'Session Time' category includes Session Time, TDD Ratio, Software Version, Tower, Status, and Status Time. The 'TDD Ratio' category includes DNS, Default Gateway, LAN Mode Status, LAN Speed Status (Mbps), LAN Status, LAN Status, Netmask, Network LAN MTU (Bytes), WAN IP Address, WLAN Status, and Wireless MAC. The 'Network' category includes Channel Width, DFS Status, DL Frame Utilization, and MCS. The interface is clean and modern, with a blue and white color scheme.

Navigation Bar: Cambium Networks | cnMaestro™ X | Administrator | SUPPORTAIRTEL

Left Sidebar: Home, Monitor and Manage, Inventory, Map, Onboard, Managed Services, Shared Settings

Main Content Area:

- Towers > Bangalore_Test**
- Breadcrumb:** Dashboard > Notifications > Configuration > Statistics > Report X > Software Update
- Sub-Navigation:** Devices (selected), Performance, Active Alarms, Alarm History, Events, Clients, Guest Access Login, Events
- Metrics (all selected):**
 - Session Time:** Session Time, TDD Ratio, Software Version, Tower, Status, Status Time
 - TDD Ratio:** DNS, Default Gateway, LAN Mode Status, LAN Speed Status (Mbps), LAN Status, LAN Status, Netmask, Network LAN MTU (Bytes), WAN IP Address, WLAN Status, Wireless MAC
 - Network:** Channel Width, DFS Status, DL Frame Utilization, MCS

Inventory and Statistics Reports

Inventory &
Statistics reports
can be export
based on Tower
group or as per
Network

Single or In Bulk Firmware upgrade

- New Firmware image need to be downloaded from Support Center and added to the cnMaestro server.
- Selected single device firmware upgrade
- Multiple device firmware upgrade as per Tower group devices and as per Network configuration template
- Only AP or Only SM can be upgraded.
- SM can be upgraded first and then AP
- AP first and then SM
- Schedule based configuration of single devices or a group of devices across the network using ePMP devices templates.

© 2020 Cambium Networks, Ltd

© 2020 Cambium Networks, Ltd

Create Configuration Templates and Apply on Single or in Multiple devices

- Configuration template can be created based on user requirement.
- Selected single configuration for creating template.
- Select Multiple configuration for creating template
- Apply selected Configuration Templates on Single Device or group of devices across the Network
- Schedule based configuration of single devices or a group of devices across the network using ePMP devices templates.

Server Backup

- Server database backup can be generate at any point of time
- Backup can be schedule Daily and Weekly
- Backup can be Restored at any point of time

Monitor Server Health

Monitor virtual machine parameters such as disk, memory, and CPU utilization through the UI.

Mail Notification based on Alarms Severity

Cambium Networks | cnMaestro™ X

Administration > Server

Monitoring **Settings** Operations Diagnostics SSL Certificates Software Images

Configure Email Server

Configure SMTP server to manage cnMaestro users pertaining to MSP and to send email notifications.

☒ Enable SMTP Server

Port*

Host*

Username

Password [Show](#)

Sender Email*

Encryption
☐ None ☒ TLS ☐ STARTTLS

☐ Ignore server certificate validation

CA Certificate [Select File](#)

[Send Test Mail](#)

Login Security Banner

Cambium Networks | cnMaestro™ X

Administration > Settings

General **Notifications** Syslog X Webhooks X Cloud X

Settings

Please ensure that SMTP server is enabled and configured.

☐ Enable email notification

Configure Email IDs to subscribe for email notifications for devices.

Email	Content Type

[Add Recipient](#)

Add Email Subscriber

☒ Active

Severity
Critical

All alarms of chosen severity or greater will be sent.

Email

Content Type
☒ HTML ☐ JSON

Ignore Notification
☒ cnPilot Home Offline
☒ ePMP SM Offline
☒ PMP SM Offline

[Add](#) [Cancel](#)

Copyright © 2015 - 2022 Cambium Networks, Ltd. All rights reserved. | Version 3.1.0-r1 | [Community](#) | [Support](#) | [Help](#) | [License](#)

Audit Logs

 Cambium Networks

cnMaestro™ X

 Home

 Monitor and Manage

 Inventory

 Map

 Onboard

 Managed Services X

 Shared Settings >

 Network Services >

 Administration >

Users

Jobs

Server

Settings

Sync Configuration

Audit Logs X

Administration > Audit Logs x

Export

Result	Time	Type	Module	Action	Source	IP Address	Description
Success	Tue Jun 21 2022 07:12:17 UTC -0...	Onboarding	Onboard	Claim	Administrator	192.168.100.209	Device onboarding initiated for device BC:E6:7C:71:42:25
Success	Tue Jun 21 2022 07:11:44 UTC -0...	Security	Administrator	Login	Administrator	192.168.100.209	Administrator logged in successfully
Success	Tue Jun 21 2022 01:40:04 UTC -...	Provisioning	Onboard	Edit	Administrator	192.168.100.209	Onboarding Key settings updated for users admin
Success	Tue Jun 21 2022 01:06:56 UTC -...	Operations	TEMPLATE	Create	Administrator	192.168.100.209	System level template Backup initiated
Success	Tue Jun 21 2022 01:06:37 UTC -...	Operations	TEMPLATE	Download	Administrator	192.168.100.209	System level template export successful
Success	Tue Jun 21 2022 00:42:55 UTC -...	Onboarding	Onboard	Claim	Administrator	192.168.100.209	Device onboarding initiated for device 58:C1:7A:B2:4D:31
Success	Mon Jun 20 2022 23:28:42 UTC...	Security	Administrator	Create	Administrator	192.168.100.209	Added new user ashwani with operator role
Success	Mon Jun 20 2022 23:13:20 UTC ...	Operations	Device	Edit	Administrator	192.168.100.209	Device details updated for device 58:C1:7A:B2:4C:B6
Success	Mon Jun 20 2022 23:13:20 UTC ...	Operations	Device	Move	Administrator	192.168.100.209	Device topology updated for 58:C1:7A:B2:4C:B6
Success	Mon Jun 20 2022 23:13:20 UTC ...	Operations	Device	Move	Administrator	192.168.100.209	Device topology updated for 58:C1:7A:B0:73:3A

Showing 1 - 10 Total: 115 10 < Previous 1 2 3 4 5 ... 12 Next >

Copyright © 2015 - 2022 Cambium Networks. All rights reserved. | Version 3.1.0-r1 | [Community](#) | [Support](#) | [Help](#) | [License](#)



Events Syslog and Audit SysLog can be Forwarded to a remote system log server.

 Cambium Networks

cnMaestro™ X



Administrator
SUPPORTAIRTEL



Administration > Settings

General

Notifications

Syslog X

Webhooks X

Cloud Connectivity

Event Syslog

Add

Enable	Name	IP/Host	Port	Event Type	Severity	New Facility	New Severity
No Data Available							

Audit Syslog

Add

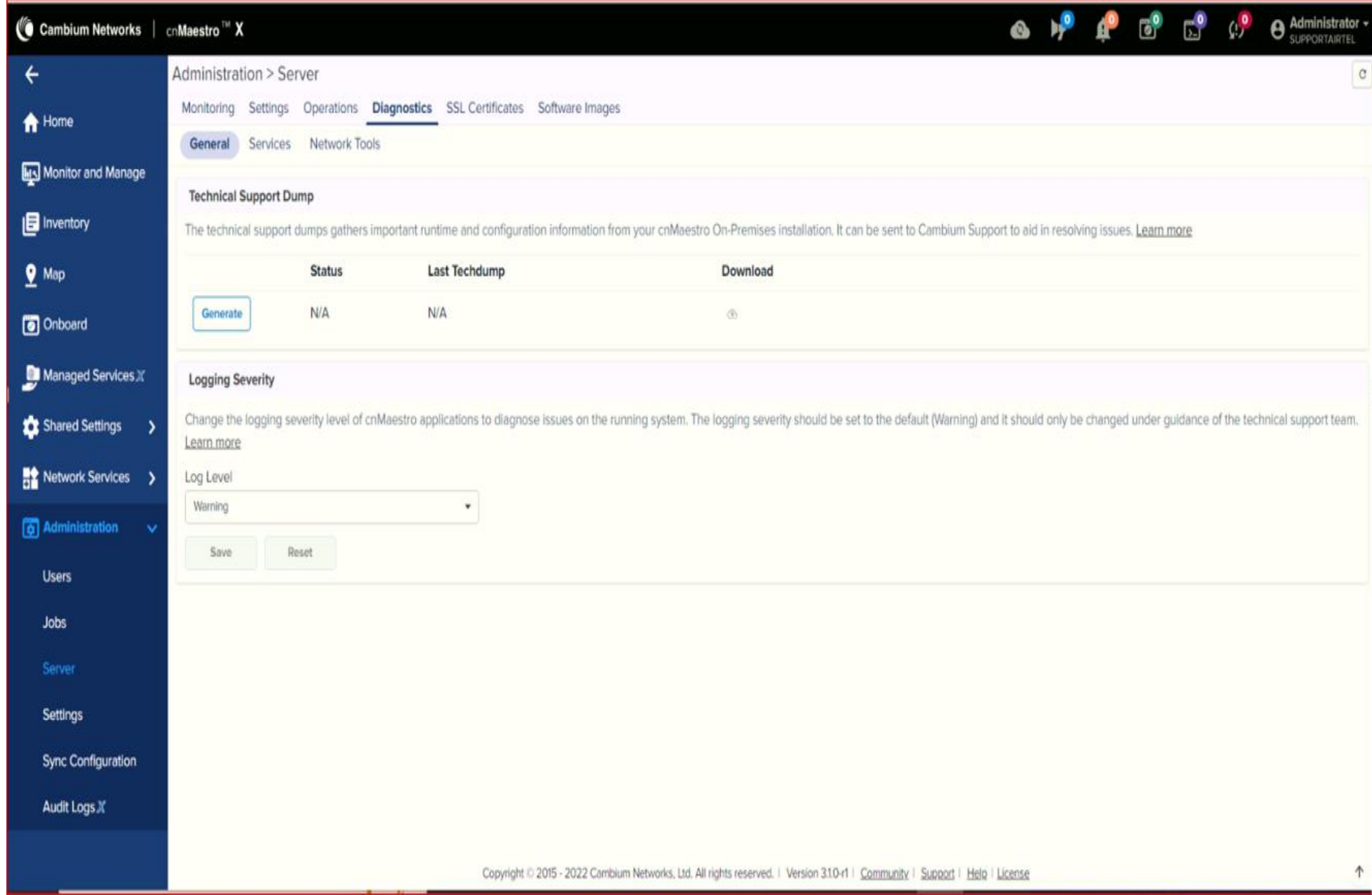
Enable	Name	IP/Host	Port	Audit Type	New Facility	New Severity
No Data Available						

Copyright © 2015 - 2022 Cambium Networks, Ltd. All rights reserved. | Version 3.1.0-r1 | [Community](#) | [Support](#) | [Help](#) | [License](#)

↑

Technical Support Dump

- In case any issue in Server, Generate Technical support dump and Download
- Sent it to Cambium Support Team to Analyse the issue and suggest resolution purpose.



The screenshot shows the Cambium Networks cnMaestro Administration interface. The left sidebar contains navigation links: Home, Monitor and Manage, Inventory, Map, Onboard, Managed Services, Shared Settings, Network Services, Administration (selected), Users, Jobs, Server, Settings, Sync Configuration, and Audit Logs. The main content area is titled 'Administration > Server' and includes tabs for Monitoring, Settings, Operations, Diagnostics (selected), SSL Certificates, and Software Images. Under the Diagnostics tab, there are sub-tabs for General, Services, and Network Tools. The 'General' sub-tab is active, displaying the 'Technical Support Dump' section. This section explains that the dump gathers runtime and configuration information and can be sent to Cambium Support. It includes a table with columns for Status, Last Techdump, and Download. The Status column has a 'Generate' button, and the Last Techdump column shows 'N/A'. The Download column has a download icon. Below this is the 'Logging Severity' section, which allows changing the logging severity level of cnMaestro applications. It includes a 'Log Level' dropdown menu set to 'Warning' and 'Save' and 'Reset' buttons. The footer contains copyright information: Copyright © 2015 - 2022 Cambium Networks, Ltd. All rights reserved. | Version 3.10-r1 | Community | Support | Help | License.

Cambium Networks | cnMaestro™ X

Administration > Server

Monitoring Settings Operations **Diagnostics** SSL Certificates Software Images

General Services Network Tools

Technical Support Dump

The technical support dumps gathers important runtime and configuration information from your cnMaestro On-Premises installation. It can be sent to Cambium Support to aid in resolving issues. [Learn more](#)

	Status	Last Techdump	Download
Generate	N/A	N/A	

Logging Severity

Change the logging severity level of cnMaestro applications to diagnose issues on the running system. The logging severity should be set to the default (Warning) and it should only be changed under guidance of the technical support team. [Learn more](#)

Log Level

Warning

[Save](#) [Reset](#)

Copyright © 2015 - 2022 Cambium Networks, Ltd. All rights reserved. | Version 3.10-r1 | [Community](#) | [Support](#) | [Help](#) | [License](#)

Questions

